

Housing market responses to the mortgage interest deduction

Sven Damen & Geert Goeyvaerts

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Abstract

This paper examines the impact of tax subsidies for homeowners on the housing market. In 2015, the Belgian regions became responsible for the tax subsidy after having been under federal control. The average present value of the subsidies decreased significantly more in Flanders than in Wallonia. We exploit this variation in a difference-in-differences design to estimate the effect of its repeal on equilibrium prices and quantities. There are two important results. Changes in the subsidies largely capitalize in the price of housing and land, while their quantities remain unaffected. These results indicate that the subsidies are largely captured by current property owners and can be gradually reduced without disrupting the market.

Keywords: Mortgage interest deduction, Housing market, Economic incidence.

Damen: Department of Accountancy and Finance, University of Antwerp. (E-mail: sven.damen@uantwerp.be) Goeyvaerts: Department of Economics, KU Leuven. (E-mail: geert.goeyvaerts@kuleuven.be)

1 Introduction

Policy-makers often seek to support the dream of many households to become homeowners. One of the most commonly used instruments to achieve this objective is the mortgage interest deduction (MID), which enables tax payers to deduct their mortgage interest payments from their personal income. However, a growing number of studies show that these tax subsidies have been largely ineffective. As a result, governments started to question the usefulness of the MID and looked at possible reforms.¹

We investigate the repeal of the mortgage payment deduction (MPD) in Belgium. The MPD allowed tax payers to deduct both the interest and principal payment up to a maximum amount. In 2014, the responsibility for the MPD was transferred to the Belgian regions, resulting in a series of reforms that reduced the average present value of the future tax subsidies substantially more in Flanders than in Wallonia.² We exploit this variation in a difference-in-differences design to obtain credible estimates of the effect of housing tax subsidies on the equilibrium price and quantity of housing and land.

There are two main findings, both with important policy implications. First, the reduction in the discounted future tax subsidies for homeowners largely capitalizes in the price of housing and land. The estimates from our preferred specifications indicate that one euro decrease in the present value lowers the price of single-family houses by one euro and the price of plots of undeveloped land by eighty cents. This nearly full capitalization implies that property owners received the largest share of the subsidies through higher property prices.

Second, the much larger reduction in the tax subsidy in Flanders did not significantly

¹The [Council of Economic Advisers \(2017\)](#) evaluate the effects of changes in the mortgage interest deduction as the Tax Cuts and Jobs Act of 2017 in the U.S., which doubled the standard deduction for households and reduced the cap on the mortgage interest deduction. Several member states of the European Union reformed or repealed their mortgage interest deduction, e.g. Ireland and Spain in 2013, Portugal in 2012, France in 2011 and Lithuania in 2009 ([Leodolter and Rutkowski, 2022](#)).

²Flanders and Wallonia are the two largest regions of Belgium. The Brussels Capital Region is the third region, but is much smaller in size since it only encompasses the city of Brussels.

decrease sales of existing houses, sales of undeveloped land nor construction permits for new houses. The insignificant effect on equilibrium quantities does not only shed light on the mechanism behind the price effects, as it indicates supply of land is considerably price-inelastic, but it also implies the MID can be gradually repealed without severely disrupting the housing market.³

We contribute to the literature that studies the effects of the mortgage interest deduction on the housing market. A large body of academic research uses calibrated quantitative models to study the impact of the MID.⁴ This paper is more related to the literature that exploits quasi-experimental variation to estimate different parameters of interest.⁵ A first set of studies exploits variation across U.S. states. [Hanson \(2012\)](#) finds that the MID increases home size but not homeownership. [Hilber and Turner \(2014\)](#) show that a higher mortgage subsidy rate decreases the probability that high income households are homeowner in markets with restrictive land use regulations, while it increases homeownership attainment in less tightly regulated markets. Lastly, [Davis \(2019\)](#) finds that a one percentage point increase in the tax rate applied to mortgage interest raises house prices by 0.8%.

Other studies have been able to look at reforms in which the tax subsidies were reduced. [Jappelli and Pistaferri \(2007\)](#) find that a reform in the early 1990s in Italy did not decrease the equilibrium amount of mortgage debt, neither at the extensive nor the intensive margin. In contrast, [Alan et al. \(2016\)](#) show that a reduction in marginal tax rates on capital income in Denmark in the late 1980s decreased interest payments. Exploiting the same reform, [Gruber et al. \(2021\)](#) confirm the negative effect on indebtedness but also show that the reform decreased home size without affecting homeownership attain-

³In 2017, the National Association of Realtors' deputy chief lobbyist argued that the MID is one of the biggest incentives to buy a home. Take it away and "obviously sales volumes will drop", he said. See Marshall-Genzer (Sep 25, 2017) "Lobbyists line up to defend the mortgage interest deduction" on marketplace.org (Accessed on May 5, 2021).

⁴See [Gervais \(2002\)](#), [Díaz and Luengo-Prado \(2008\)](#), [Chambers et al. \(2009\)](#), [Floetotto et al. \(2016\)](#), [Sommer and Sullivan \(2018\)](#), and [Karlman et al. \(2020\)](#) among others.

⁵See [Gruber et al. \(2021\)](#) for a discussion of earlier empirical work such as [Glaeser and Shapiro \(2003\)](#).

ment. In addition, the authors find that house prices declined 12% relative to a synthetic control country in the first year after the reform.

Our results confirm some of these previous findings, but we also make a couple of important contributions. First, while the previous estimates of the effect on house prices suggest buyers did not benefit from the tax subsidies, an accurate assessment of the incidence requires estimates that allow us to gauge the extent to which the subsidies capitalize in the price of the subsidized good. We present the first estimates of the capitalization rate of the mortgage interest deduction in house prices.⁶ Second, we present the first estimates of the effect on land prices. These results highlight that the owners of the most price-inelastic production factor were the main beneficiaries of the subsidies. Third, our estimates of the effect on the number of sales and construction permits are the first that indicate the equilibrium quantity of housing does not change at the extensive margin. They confirm the null effect on homeownership and at the same time complement the finding of a significant change at the intensive margin.

This study is organised as follows. Section 2 provides a theoretical framework which illustrates how tax subsidies affect the equilibrium prices and quantities of housing and land. Section 3 describes the MPD and the reforms at the regional level. Section 4 presents the data that we use in the empirical analysis. Section 5 describes our empirical strategy. Section 6 presents the empirical results on prices and quantities. We conclude in Section 7.

⁶[Berger et al. \(2000\)](#) show that interest rate subsidies in Sweden largely capitalize into the price of the houses to which the subsidized loan is tied. However, their analysis should be considered as a test of the efficiency of the housing market rather than a measurement of the economic incidence of subsidies for homeowners. Their results confirm that the price of similar products should reflect differences in taxation. They do not tell us whether the price of the subsidized dwellings would be lower or the price of the non-subsidized houses would be higher in case the subsidized loans would not have been introduced.

2 Theoretical framework

In this section, we present a simple theoretical model to show how tax subsidies for homeowners affect equilibrium prices and quantities of housing and land. As in [Glaeser et al. \(2012\)](#) each period t , K potential residents compare the net discounted cost (NDC) of buying a house against the discounted future value of utility from living in the house for an infinite horizon. The NDC of buying a house equals the sum of the down payment and the present value of mortgage payments minus the present value of the future tax subsidies⁷:

$$dP_{h,t} + \sum_{n=1}^N \frac{(1-d)P_{h,t}}{a(N, i_t)} \frac{1}{(1+i_t)^n} - \sum_{n=1}^N \frac{s_t}{(1+i_t)^n} = P_{h,t} - S_t \quad (1)$$

Where d equals the down payment, $P_{h,t}$ is the price of a house, a is the annuity formula which is a function of the term to maturity of the mortgage N and the mortgage interest rate i_t , s_t is the annual tax subsidy and S_t is the present value of the annual tax subsidies.

A household receives a euro-denominated utility flow from owning a house of $A_{i,t} = (1+g)^t A_i$, where A_i is the household-specific utility from owning the house and g the yearly growth rate of the euro-denominated utility flow. $\bar{A}$ denotes the utility for the marginal buyer, who is indifferent between owning and the alternatives. Assume A_i follows a Pareto distribution with parameter $1/\gamma$ such that the probability that A_i is larger than $\bar{A}$ equals $\bar{A}^{-1/\gamma}$. The demand for housing units ($Q_{h,t}^D$) is equal to $K\bar{A}^{-1/\gamma}$. Rewriting gives the household-specific taste from owning a house for the marginal buyer: $\bar{A} = K\gamma(Q_{h,t}^D)^{-\gamma}$. The marginal buyer at time t compares the discounted future value of utility to the NDC of buying the house for an infinite horizon such that

$$\frac{1}{i_t - g} \left(\frac{K}{Q_{h,t}^D} \right)^\gamma = P_{h,t} - S_t. \quad (2)$$

Rewriting Equation 2 gives us the demand for houses as a function of the present value

⁷We assume that the discount rate is higher than the appreciation rate of house prices such that the resale value should not be included in the NDC of buying with an infinite horizon.

of future tax subsidies:

$$Q_{h,t}^D = K \left[(i_t - g) (P_{h,t} - S_t) \right]^{-\frac{1}{\gamma}} \quad (3)$$

Taking the derivative of (3) with respect to S_t shows that demand shifts upwards when the present value of future subsidies increases.

Total supply of houses exists of existing and newly built houses. First, each period a fixed fraction of the existing housing stock is supplied inelastically to the market: $Q_{h,t}^E = c.H_t$.⁸ Second, developers supply new houses by building a house on a plot of undeveloped land. They are price-taking agents in a perfectly competitive market and maximize the following profit function:

$$\pi = (P_{h,t} - P_{l,t} - P_k) Q_{h,t}^N \quad (4)$$

In which $P_{l,t}$ is the price of a plot of undeveloped land, P_k is the price of the construction and $Q_{h,t}^N$ the quantity of newly built houses. We assume that the supply of non-land construction inputs is perfectly price-elastic. The price of the construction therefore equals its marginal cost, which we assume is fixed over time. The supply of plots of undeveloped land ($Q_{l,t}^S$) equals $LP_{l,t}^\psi$ such that its price-elasticity is given by ψ . Since developers make zero profits, the price of a house and plot of land are respectively

$$P_{h,t} = P_{l,t} + P_k \quad \text{and} \quad P_{l,t} = P_{h,t} - P_k. \quad (5)$$

Each period, demand for houses should equal the supply of houses:

$$Q_{h,t}^D = Q_{h,t}^E + Q_{h,t}^N = c.H_t + LP_{l,t}^\psi. \quad (6)$$

Assuming the supply of land is perfectly price-inelastic, i.e. $\psi = 0$, we substitute (3) in (6) and solve for $P_{h,t}$ to get the equilibrium price of housing as a function of the present

⁸Note that we abstract from depreciation and redevelopment in this model.

value of the annual tax subsidies:

$$P_{h,t} = \left(\frac{K}{c \cdot H_t + L} \right)^\gamma \frac{1}{(i - g)} + S_t. \quad (7)$$

Taking the derivative of (7) with respect to S_t shows that a one euro increase in the present value of future tax subsidies increases house prices by one euro. In other words, the present value of future tax subsidies capitalizes completely in house prices. We can further substitute (5) into (7) and solve for $P_{l,t}$ to find that a one euro increase in the present value of future subsidies leads to an increase in land prices of one euro.

The following four equalities summarize the main predictions of this model:

$$i) \quad \frac{\partial P_{h,t}}{\partial S_t} = 1 \quad ii) \quad \frac{\partial P_{l,t}}{\partial S_t} = 1 \quad iii) \quad \frac{\partial Q_{h,t}^*}{\partial S_t} = 0 \quad iv) \quad \frac{\partial Q_{l,t}^*}{\partial S_t} = 0 \quad (8)$$

Together they state that one euro increase in the present value of future housing subsidies will raise house and land prices by one euro without an effect on the equilibrium quantity of new houses and plots of undeveloped land.

The predictions in (8) hinge on the assumption of a perfect price-inelastic supply of land for residential development. To our knowledge, there does not exist an estimate of the price elasticity of the supply of land. However, [Caldera Sánchez and Johansson \(2013\)](#) estimate that the price-elasticity of the supply of new houses in Belgium is 0.315 in the long-run and 0.244 in the short-run. Although the latter estimate is not significantly different from zero. The assumption of a perfectly inelastic supply of land thus seems a reasonable starting point given the existing evidence. If the price-elasticities of the supply of land and newly built houses is larger than zero, changes in the present value of future tax subsidies will not completely capitalize in prices and equilibrium quantities will increase. In the next section, we discuss the variation in housing tax subsidies that we exploit to test the predictions in (8).

3 Institutional background

Belgium tried to stimulate homeownership using a system similar to the mortgage interest deduction. The Belgian version of the MID was introduced in 2005.⁹ It allowed tax payers to deduct annual mortgage payments from their taxable income.¹⁰ We will therefore refer to this tax subsidy as the mortgage payment deduction (MPD).

The size of the MPD depended on the size of the mortgage payment tax payers deducted and the tax rate at which it was deducted. The size of the mortgage payment that could be deducted was capped. In 2005, tax payers could deduct up to 1,870 Euro. They were allowed to deduct an additional 620 Euro during the first ten years of the mortgage, bringing the total deductible amount to 2,490 Euro.¹¹ These caps were adjusted to inflation every year. Table 1 reports the caps from 2005 to 2014. Aggregate data on the use of the MPD in Belgium in combination with survey data from the European Household Finance and Consumption Survey (HFCS) suggest that these caps were binding for most home buyers.¹²

The mortgage payment was deducted from the taxable personal income, which mainly consists of labour income for employees or earnings in case the tax payer is self-employed.¹³ There were five tax brackets in Belgium over the period from 2005 to 2014. Income in these brackets was taxed at the following rates: i) 25%, ii) 30%, iii) 40%, iv) 45% and v) 50%.¹⁴ The use of the MPD was concentrated in the upper three tax brackets. The population of tax payers was distributed across the five tax brackets as follows: i) 23,4%, ii) 12,1%, iii) 26,5%, iv) 26,9% and v) 11,2%.¹⁵ The population of tax payers that deduct a mortgage payment were distributed as follows: i) 8,9%, ii) 6,6%, iii) 20,5%, iv) 46,3%

⁹It replaced a similar but less beneficial tax subsidy.

¹⁰The mortgage payment includes both the interest payment as well as the principal.

¹¹The MPD was a tax subsidy for an individual tax payer. If a couple wanted to enjoy the maximum subsidy, the annual mortgage payment had to be twice as large as the cap.

¹²We present and discuss these data in Section A.1 in the Appendix.

¹³Capital income is taxed at separate fixed rates.

¹⁴Table A.5 in the Appendix reports the income thresholds on the five tax brackets.

¹⁵Table A.7 in the Online Appendix provides the distribution of tax payers across the five tax brackets for the period of 2005 to 2013 for Belgium, Flanders and Wallonia.

and v) 17,7%.¹⁶ The average marginal tax rate at which the mortgage payments were deducted fluctuated around 42% over the pre-reform period.¹⁷

Table 1. The caps on the federal mortgage payment deduction, 2005 to 2014

	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014
Maximum first 10 years	2,490	2,560	2,600	2,650	2,770	2,770	2,830	2,930	3,010	3,040
Maximum after 10 years	1,870	1,920	1,950	1,990	2,080	2,080	2,120	2,200	2,260	2,280
Add-on first 10 years	620	640	650	660	690	690	710	730	750	760

Notes: This table reports the caps on the mortgage payment that could be deducted.

When the regions became responsible for the MPD, each passed different reforms. Flanders made two important adjustments to the MPD in 2015.¹⁸ First, the mortgage payment deduction was turned into a mortgage payment reduction at a fixed rate of 40%.¹⁹ Second, the cap on the mortgage payment was reduced by a third, from 2,280 euro to 1,520 euro. Since the add-on was not reduced, the maximum deductible mortgage payment during the first ten years decreased by a quarter to 2,280 euro.

Wallonia implemented two consecutive reforms. In their first reform in 2015, they also turned the MPD into a tax reduction at 40% but they did not reduce the caps. In their second reform in 2016, they replaced the MPD by a novel tax reduction called the 'Chèque Habitat' (CH). All owner-occupiers with a mortgage and an annual taxable income of less than 21,000 euro receive an annual tax credit of 1,520 euro.²⁰ If their taxable income is higher than 21,000 euro then the credit is reduced by 1.275% of the difference between the annual taxable income and the threshold. Tax payers with a taxable income above 81,000 euro did not benefit from the Chèque Habitat in 2016.²¹

¹⁶These shares were relatively constant over time. Table A.8 in the Online Appendix provides the distribution of tax payers that deduct a mortgage payments across the five tax brackets for the period of 2005 to 2013 for Belgium, Flanders and Wallonia.

¹⁷See Table A.4 in the Online Appendix.

¹⁸In addition to these two adjustments, the caps were no longer adjusted to inflation.

¹⁹Belgian regions are not allowed to introduce tax *deductions* but only tax *reductions* and *credits*. This is to prevent them from reducing the tax base of the federal government.

²⁰To be eligible for the credit, tax payers in Wallonia have to live in the house for which they hold a mortgage, i.e. be an owner-occupier. Tax payers who own more than one property - whether developed or undeveloped - can benefit from a different but less beneficial tax reduction.

²¹The income threshold is adjusted to inflation each year.

Wallonia also restricted the use of the CH in time. The tax credit is reduced by half if a tax payer enjoyed it more than ten years and it is completely halted after twenty years. Note that this is not the case for the federal nor the Flemish MPD. While the cap on the mortgage payment decreases by about a third ten years after the mortgage is closed (see Table 1), tax payers can deduct a mortgage payment as long as all eligibility criteria are satisfied.

The Brussels Capital Region also made use of its new competence and repealed the MPD in 2017. However, it also drastically increased the exempt in the transaction tax to compensate for the repeal of the MPD. We will exclude Brussels from the analysis since it will be impossible to separately identify the effects of these two policy changes. Furthermore, we limit the analysis to the first quarter of 2018 since Flanders lowered the transaction tax in June 2018.

To illustrate how the reforms affected the size of the tax subsidies, we computed the present value (PV) of the subsidies for the average couple of home buyers in Flanders and Wallonia.²² Figure 1 plots the average PV for Flanders (solid) and Wallonia (dashed) together with the difference (dotted) from 2005M1 to 2018M3.²³ The difference in the average PV's decreased by 7,408 euro because of the reforms, from an average of 738 euro before the reforms to an average of -6,670 euro after the reforms.²⁴ We will exploit this sharp decrease in the difference to identify the effect of the housing subsidies on prices and equilibrium quantities of housing and land. In the next section, we discuss the data that we use to measure prices and quantities.

²²See Section A.2 in the Online Appendix for a detailed discussion of how we computed the average PV.

²³The increase in the average PV's over the pre-reform period is largely driven by the decrease in the average mortgage interest rate, which we use as the discount rate, and to a lesser extent by the adjustment of the caps to inflation.

²⁴The pre-reform difference of 738 euro can be attributed to the 1.3 percentage points higher average marginal tax rate in Flanders.

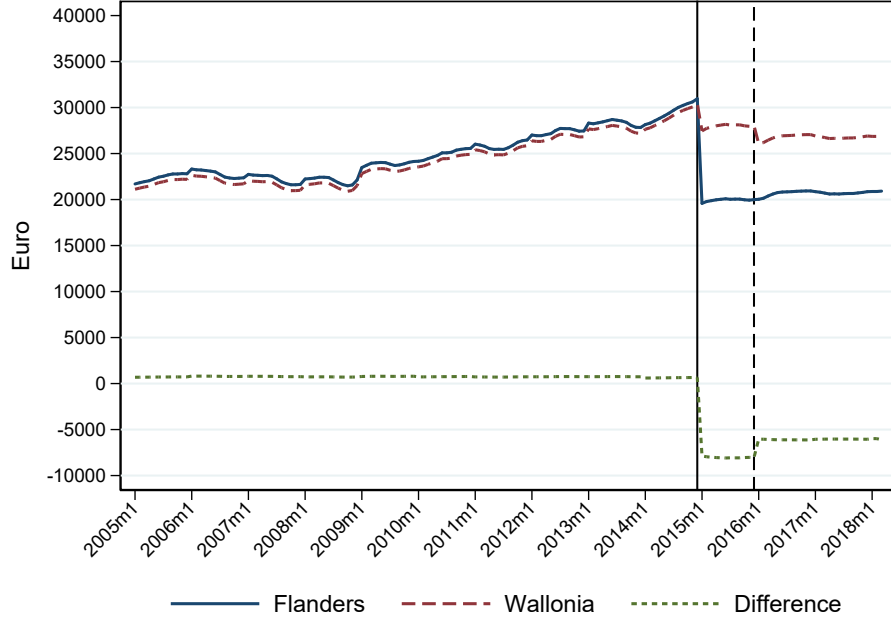


Fig 1. The average present value of the stream of future housing tax subsidies in Flanders and Wallonia

Notes: This figure plots the weighted average present value of future housing tax subsidies in Flanders (solid) and Wallonia (dashed) together with the difference (dotted). The vertical lines indicate the last month before the two regional reforms: i) the adjustments to the MPD in 2015 (solid) and ii) the introduction of the Chèque Habitat in Wallonia in 2016 (dashed). We discuss the computation of the present value of future tax subsidies in detail in Section A.2 in the Online Appendix.

4 Data

We use two data sets to test the theoretical predictions in Section 2. The first data set is the universe of sales of existing properties in Flanders and Wallonia starting from 2009, which we obtained from the General Administration of Patrimonium Documentation (GAPD).²⁵ We use these data to estimate the effect of housing tax subsidies on the price of houses ($P_{h,t}$), the price of plots of undeveloped land ($P_{l,t}$), sales of existing houses ($Q_{h,t}^E$) and sales of plots ($Q_{l,t}^*$).²⁶ The second data set is the number of construction permits for

²⁵We received the universe of transactions of real estate in Belgium. We removed all transactions that are not regular at arm's length sales, such as inheritances, and all transactions of real estate located in the Brussels Capital Region.

²⁶The sales data do not contain sales of newly built houses because newly built houses are subject to value-added tax (VAT) and sales of existing houses and plots of land are subject to the transaction tax.

single family houses starting from 1997, which are made publicly available by Statistics Belgium (Statbel). We use construction permits as a proxy for the equilibrium quantity of newly built houses ($Q_{h,t}^N$).

4.1 Sales data

The information in the sales data comes from two sources: i) the notarial deed and ii) the land register. The deed provides information on the sale itself, like the date and price, and it provides some information on the property, like the type of property and the size of the plot. The land register also provides the type of property and the size of the plot in addition to several building characteristics in case the plot is developed like the type of construction (i.e. terraced, semi-detached or detached), construction year, square footage, the number habitable rooms, bathrooms and garages. Furthermore, the register provides a unique identification key for each plot and the address of the building if the plot is developed.²⁷

We converted the data into a data set consisting of sales of houses and plots of undeveloped land. Initially, we removed all sales with missing prices, as well as sales of multiple properties. Sales of multiple properties usually have a single price, which makes it impossible to determine the price of each individual property. We then selected sales of single family houses and plots of undeveloped land for residential development based on the description in the notarial deed.²⁸ We removed houses with missing addresses, construction types or construction years. The resulting data set contains 582,208 sales of existing single-family houses and 66,813 sales undeveloped plots for residential development.

Figure 2 plots the quarterly number of sales of single-family houses (panel a) and

²⁷The address of multifamily housing units often lack the housing units' number. Therefore, we are unable to uniquely identify the different units in a building. Since tracking units over time will be an important part of our identification strategy, we exclude multifamily housing from the analyses.

²⁸A building is considered a single family house if the deed describes it as a "*Woonhuis met aanhorigheden op en met grond*" or "*Villa met aanhorigheden op en met grond*", and an undeveloped plot is considered as land for residential redevelopment if the deed describes it as "*Bouwgrond*".

plots of undeveloped land (panel b) for Flanders and Wallonia. There are a couple of features worth highlighting. First, the average number of sales of houses and plots are respectively two and three times higher in Flanders than in Wallonia. This large difference is the result of several factors of which population size is probably the most important driver.²⁹

Second, sales of houses and plots in Flanders increased sharply in the fourth quarter of 2014 after which it decreased and slowly recovered over the course of 2015. The reform of the MPD in Flanders was announced in July 2014. The peak and subsequent trough suggest households moved their decision to sell or buy forward in time. The number of households who anticipated the reforms seems to have been smaller in Wallonia. The estimates of the effects on sales will be biased downwards if the anticipation effect was larger in Flanders than in Wallonia. To address this potential bias, we exclude the fourth quarter of 2014 to the first quarter of 2016 when we estimate the effect on sales.

Third, sales did not decrease after the reforms. Instead, they seem to have increased and seemingly more so in Flanders than in Wallonia. The average quarterly number of sales of houses increased 22.9% in Flanders, whereas it increased 15.4% in Wallonia.³⁰ The difference in growth rates is even larger for sales of undeveloped plots. The post-reform average number of sales is about 9% higher than the pre-reform average in Flanders, while it is 15% lower in Wallonia.

²⁹On January 1 2014, Flanders had a population of 6.4 million inhabitants whereas Wallonia had a population of 3.5 million inhabitants.

³⁰Specifically, the average quarterly number of sales of houses in Flanders increased from 9,841 before 2014Q4 to 12,091 after 2016Q1. The average quarterly number of sales of houses in Wallonia increased from 5,317 to 6,135.

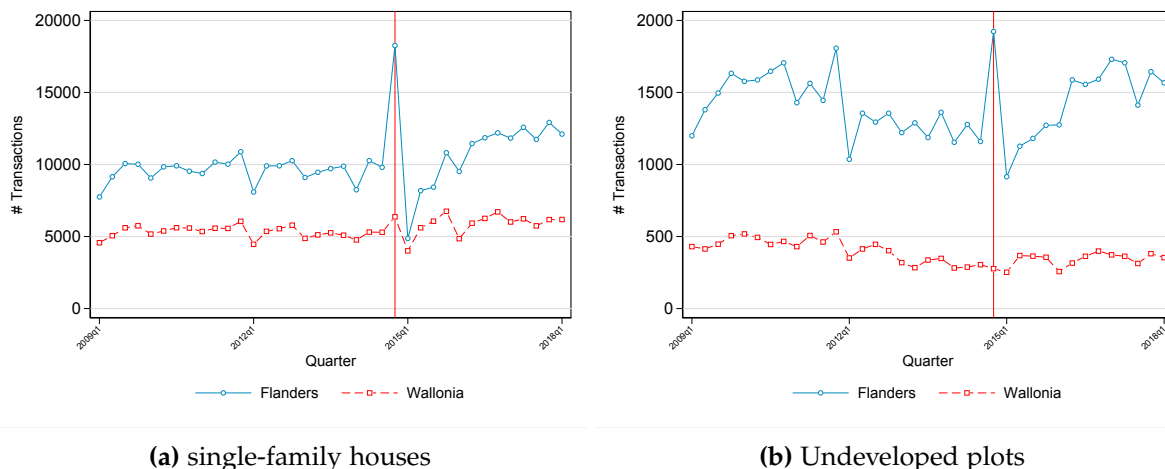


Fig 2. Sales of single-family houses and undeveloped plots over time

Notes: This figure plots the number of sales of single-family houses (panel a) and plots of undeveloped land (panel b) for Flanders (circles/solid) and Wallonia (squares/solid line) per quarter from 2009Q1 to 2018Q1. The (red) vertical line indicates 2014Q4, the last quarter before the reforms of the housing tax subsidies.

Sources: sales data from GAPD, authors' calculations.

To estimate the effect on house and land prices, we remove outliers in the two samples. We remove houses with a price, plot size or square footage below the 1st or above the 99th percentile. Furthermore, we drop houses with more than 3 garages or 10 habitable rooms as well as houses that have been sold more than 10 times over the sample period. Removing these outliers reduces the sample to 548,054 sales.

Before we remove outliers in the sample of undeveloped plots, we restrict the sample to plots that are described as land for residential development by both the deed and the land register. The GAPD changed the way they register sales of split plots in 2015. Before 2015, the GAPD would first register the sales of split plots before adjusting the information in the land register.³¹ Since the information from the land register is automatically added when a sale is registered, sales of split plots contain information on the original plot.³² In contrast, the information in the deed is at the discretion of the notary, who can provide a correct description of the split plots. However, while the way

³¹Since 2015, the GAPD first adjusts the information in the land register before they register the sales.

³²For example, when a large plot of agricultural land is subdivided into smaller plots for residential development, the sales of these plots will contain the information on the large agricultural plot.

in which the GAPD registered split plots before 2015 makes the information from the deed more reliable on average, not all notarial deeds will be error-free. There are 35,937 sales of undeveloped plots which both the deed and the register describe as land for residential development. Removing sales of plots with a price or size below the 1st or above the 99th percentile further reduces the sample to 34,706 sales.

Figure 3 plots the average price of single-family houses (panel a) and plots of undeveloped land (panel b). There are again a couple of features worth highlighting. First, there is a large difference in the regional average prices of both houses and plots. The descriptives in Panel A of Table B.1 in the Online Appendix show that the average price of houses sold in Flanders is 40% higher than the average price of houses sold in Wallonia.³³ This large difference in average prices might be the result of a difference in the quality of the construction. Houses sold in Flanders were, on average, a bit larger and more than 20 years younger. However, the descriptives in Panel B of Table B.1 suggest that a large part of the difference is driven by the difference in the average price of land. The average price per square meter of undeveloped plots sold in Flanders is more than three times higher than the average price per square meter of plots sold in Wallonia. We discuss how we account for differences in prices and quality in the next section.

Second, average prices seem to be on a fairly stable positive trend over the whole sample period. There is no sign of an increase in prices due to anticipation of the reforms nor is there any sign of a decrease in prices after the reforms.

³³The first column in Table B.2 in the Online Appendix presents the difference in the average price and building characteristics together with the p-value of a standard t-test. The difference of 66,500 euro is significantly different from zero.

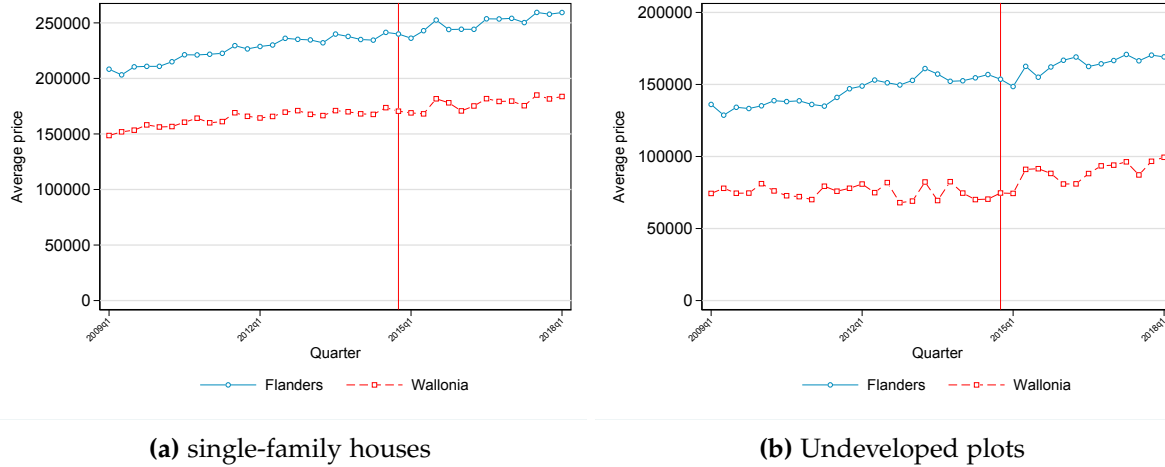


Fig 3. The average price of single-family houses and undeveloped plots over time

Notes: This figure plots the average price of single-family houses (Panel a) and plots of undeveloped land (Panel b) for Flanders (circles/solid line) and Wallonia (squares/dashed line) per quarter from 2009Q1 to 2018Q1. The (red) vertical line indicates 2014Q4, the last quarter before the reforms of the housing tax subsidies.

Sources: sales data from GAPD, authors' calculations.

4.2 Permit data

In the theoretical model, the equilibrium number of newly built houses is one-to-one related to the equilibrium number of sales of undeveloped plots. If this assumption holds in practice, the effect on sales of undeveloped plots will be the same as the effect on the number of newly built houses. In reality, however, developers do not immediately develop all the land they buy and they can develop land they already own for several years. When the demand for houses decreases due to a reduction in housing subsidies, the number of newly built houses might decrease more than the number of sales of undeveloped plots or even the other way around. We therefore also want to estimate the effect on the number of newly built houses. Since a permit is the final stage before the start of the construction, we consider the number of permits as a good proxy for the equilibrium quantity of newly build houses. Figure 4 plots the quarterly number of construction permits for single-family houses from 1997Q1 to 2018Q1.

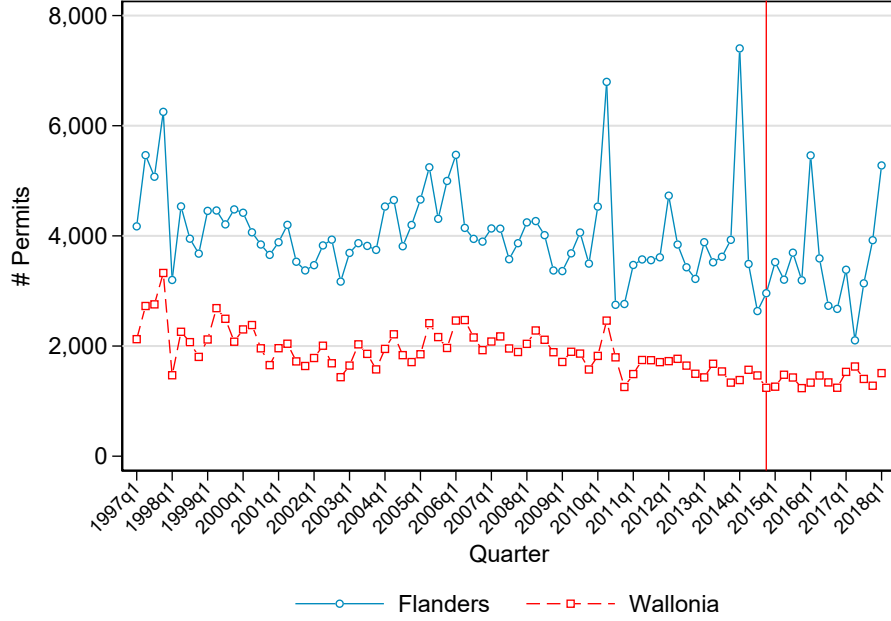


Fig 4. The number of construction permits for single-family houses over time

Notes: This figure plots the number of construction permits for single-family houses in Flanders (circles/solid line) and Wallonia (squares/dashes line) per quarter from 1997Q1 to 2018Q1. The (red) vertical line indicates 2014Q4, the last quarter before the reforms of the housing tax subsidies.

Sources: permit data from Statbel, authors' calculations.

5 Research design

The goal is to credibly estimate the effect of a change in present value of future housing tax subsidies on the price and equilibrium quantity of housing and land. As Figure 1 showed, the average PV decreased substantially more in Flanders than in Wallonia after 2014, while it was about the same before the reforms. We exploit this sharp change in the difference between the average PV's in a difference-in-differences (DD) design. The DD design compares the change in prices and quantities over time in Flanders to the change in prices and quantities in Wallonia. We estimate the following two-way fixed effects model:

$$Y_{r,t} = \beta \times \text{Flanders}_r \times \text{Post}_t + \alpha_r \times \text{Flanders}_r + \delta_t + \varepsilon_{r,t} \quad (9)$$

In which $Y_{r,t}$ indicates the outcome for region r and quarter t .³⁴ $Flanders_r$ is a dummy variable which takes the value of one if region r is Flanders. $Post_t$ indicates whether quarter t comes after the fourth quarter of 2014, i.e. the last quarter before the reforms in both regions, and δ_t is the quarter fixed effect. The coefficient of interest is β .

Callaway et al. (2021) show that under the set of assumptions we discuss below, $\hat{\beta}$ can be interpreted as the average treatment effect (ATE) of dose d . In a setup in which two (groups of) units are treated over the whole sample period, d should be computed as follows:

$$d = (d_{F,post} - d_{F,pre}) - (d_{W,post} - d_{W,pre}), \quad (10)$$

in which $d_{r,pre}$ is the average dose in region r over the pre-reform period and $d_{r,post}$ the average dose in region r over the post-reform period. The regional average PV's we computed in Section 3 indicate $\hat{\beta}$ should capture the effect of a reduction of 7,286.26 euro.³⁵

The first assumption we have to make to interpret $\hat{\beta}$ as the ATE(d) is random sampling. This assumption states that the observed data is panel data that is independent and identically distributed. To estimate the effect on the equilibrium quantities of housing and land, we can simply use the number of sales and permits as the outcome. In contrast, the average price of houses and plots of land are not a suitable outcome for two reasons. First, houses nor plots are randomly sold. The reforms of the subsidies might not only affect the price of houses and plots but also which houses and plots were sold.³⁶ Second, the average quality of the housing stock in Flanders and Wallonia might change differently over time.³⁷ We need to adjust prices for differences in quality otherwise β

³⁴We estimate the effect on prices and equilibrium quantities using data at the regional level since we do not observe any socio-economic characteristics of buyers nor construction permit applicants.

³⁵ $d = (20,479.51 - 26,651.41) - (27,149.82 - 26,035.46) = -7,286.26$.

³⁶Panel A in Table B.3 in the Online Appendix shows that the average observed quality of houses sold in 2014Q4 significantly differs from the average quality of houses sold in 2015Q1.

³⁷For example, Flanders introduced energy efficiency standards for newly built houses in 2006 while Wallonia waited until 2010. Panel B in Table B.3 in the Online Appendix shows that the average observed quality of houses sold before the reforms significantly differs from the average quality of houses sold after the reforms.

will capture the difference in the change in the average quality in addition to the effect of the change in the average present value of future housing subsidies.

Two methods are commonly used to adjust prices for changes in the composition of sales. The first method, referred to as the hedonic pricing method, regresses prices on observed characteristics of the plot:

$$P_{i,t} = \lambda' X_{i,t} + \alpha_l + \delta_t + \varepsilon_{i,t}, \quad (11)$$

With the vector $X_{i,t}$ containing the observed characteristics of plot i sold in period t and α_l some location fixed effect. The second method, referred to as the repeat sales method, regresses prices on plot fixed effects:

$$P_{i,t} = \alpha_i + \delta_t + \varepsilon_{i,t}, \quad (12)$$

with α_i the plot fixed effect. The estimates of δ_t can be interpreted as the average change in quality-adjusted prices and will be denoted $\Delta P_{r,t}$ when we use them as the dependent variable.³⁸

We can use the hedonic pricing method to estimate average changes in quality-adjusted prices of both houses and land. Equation 11 is estimated using the full sample of sales, except for the outliers we removed as described in Section 4. For houses, the vector $X_{i,t}$ includes all the characteristics listed in Table B.1 in the Online Appendix and the α_l captures the fixed effect of the so-called cadastral section in which the house is located.³⁹ For undeveloped plots, $X_{i,t}$ only includes its size and α_l are also cadastral section fixed effects.

Due to the small number of repeat sales of undeveloped plots, we can only use the repeat sales method to estimate average changes in quality-adjusted house prices. Equation 12 is estimated using a sample of 99,069 sales of 47,671 individual houses.

³⁸Both Equation 11 and 12 are estimated separately for Flanders and Wallonia.

³⁹There are 3,061 cadastral sections in Belgium.

The average house in the repeat sales sample of Flanders is 7.5% less expensive than the average house in the full sample.⁴⁰ The differences in the average characteristics in Table B.2 in the Online Appendix suggest that the difference in the average prices is the result of a difference in the average quality. The same holds for Wallonia, but both the difference in the average price and the average quality is smaller.

Choosing between the two methods entails a trade-off between external validity and potential bias. The hedonic pricing method allows us to use the whole sample of sales, but it only adjusts for changes in the observed quality and assumes all of the observed characteristics affect prices linearly. The repeat sales method, on the other hand, adjusts for changes in both the observed and unobserved quality but it assumes the quality of individual houses and plots stays fixed over time and restricts the estimation sample to plots for which we observe more than one sale. We will report the results using the repeat sales method in the main text and the results using the hedonic pricing method in the Online Appendix.

Figure 5 plots the average change in quality-adjusted prices for single family houses in Panel (a) and for plots of undeveloped land in Panel (b). There are two takeaways. The first takeaway concerns the assumption of no anticipation. The plots in Figure 2 showed that sales of houses and plots sharply increased in Flanders in the last quarter before the reforms. This suggests households in Flanders anticipated the reforms. There is, however, no sign of a spike in prices. The anticipation to the reforms might have affected the composition of what has been sold, but it did not affect prices.

The second takeaway relates to the strong parallel trends assumption, which states that all dose groups would have had the same path of potential outcomes at every dose. It is clear that this assumption is violated as the quality-adjusted price of housing and land grew at a higher rate in Flanders than in Wallonia before the reforms. The higher growth rate in Flanders will bias the estimates of β upwards if left unaccounted for. It

⁴⁰The last two columns in Table B.1 in the Online Appendix present the summary statistics for the repeat sales sample.

raises two important questions.

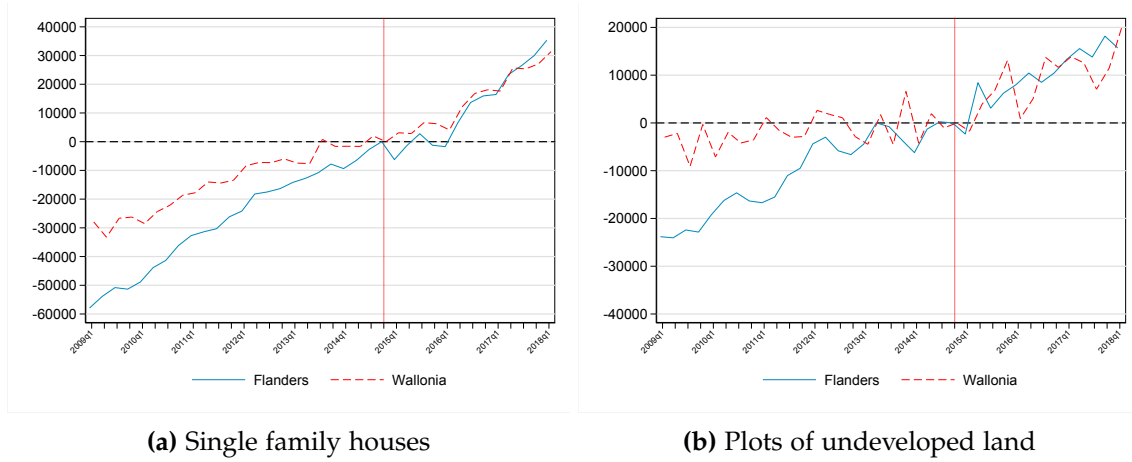


Fig 5. The average change in quality-adjusted prices

Notes: This figure plots the estimates of δ_t in Equation 12 for single family houses in panel (a) and the estimates of δ_t in Equation 11 for plots of undeveloped land in panel (b). The vertical line indicates the reference period 2014Q4, the last quarter before the start of the reforms. Panel (a) in Figure 6 in the Online Appendix plots the estimates of δ_t in Equation 11 for single family houses..

Sources: sales data from GAPD, authors' calculations.

First, why did house and land prices grew at a higher rate in Flanders? The tax subsidies did not vary across the two regions before the reforms nor did important financial variables like the mortgage interest rate. Household income, on the other hand, is an important driver of house prices whose trend did vary across the two regions.⁴¹ Panel (a) in Figure 6 plots the change in the amount of money an average household can borrow using 30% of their income.⁴² Comparing the trends in the borrowing capacity of households to the trends in prices in Panel (b) of Figure 6 and both panels in Figure 5 suggests the difference in the pre-trends is largely the result of the difference in the trends in income.

⁴¹See Duca et al. (2021) for a detailed discussion of the drivers of house price cycles.

⁴²The borrowing capacity is computed using regional GDP per capita as a proxy for income, the average mortgage maturity from the Household Finance and Consumption Survey (HFCS) which is 22.3 years and the average mortgage interest rate from the Mortgage Interest Rate (MIR) survey from the National Bank of Belgium (NBB). See Damen et al. (2016) for an analysis of the relationship between house prices and a slightly different measure of the 'ability-to-pay' (ATP).

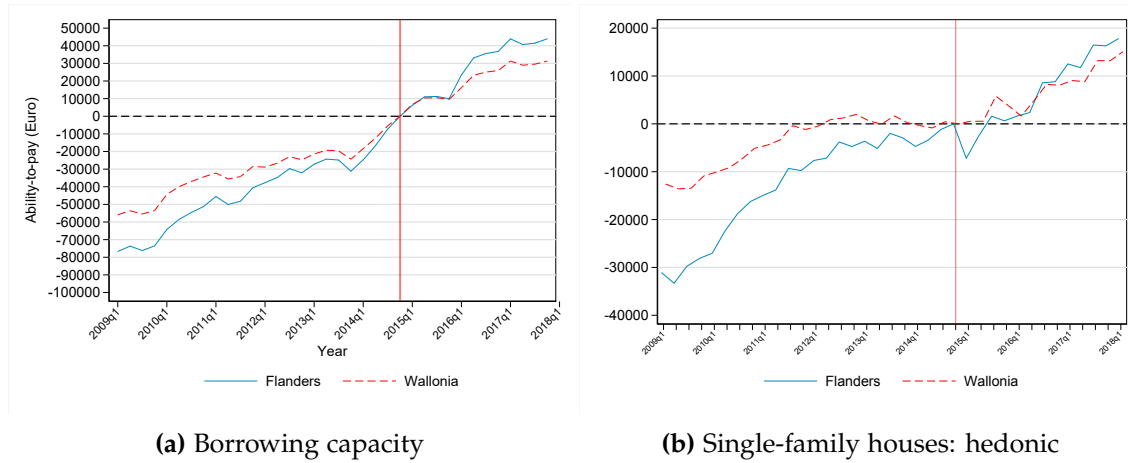


Fig 6. The change in the average borrowing capacity of households vs. the average change in quality-adjusted house prices

Notes: Panel (a) plots the change in the borrowing capacity of a household. The vertical (red) line indicates the reference period 2014Q4, the last quarter before the reforms. Panel (b) plots the average change in quality-adjusted house prices estimated using the hedonic pricing method in Equation 11. *Sources:* sales data from GAPD, authors' calculations.

Second, how can we best address the difference in the pre-reform trends? One way to deal with the pre-trends would be to use the synthetic difference-in-differences method developed by [Arkhangelsky et al. \(2021\)](#). However, the limited number of sales in some municipalities keeps us from treating municipalities as the unit of observation and we are therefore unable to apply this method. Instead, we adjust prices in Wallonia by subtracting a linear region-specific time trend which we estimate using only the pre-reform sample.⁴³ Figure 7 plots the average change in quality-adjusted prices for Flanders (solid) and the trend-adjusted average change in quality-adjusted prices for Wallonia (dash).⁴⁴ The latter closely matches the former before the reforms. After 2014, however, quality-adjusted prices in Flanders seem to have decreased relative to the trend- and quality-adjusted prices in Wallonia.

⁴³[Goodman-Bacon \(2021\)](#) points out that this procedure is preferable to directly including a unit-specific time trend in the difference-in-differences specification since it might over-control for time-varying treatment effects (p. 272).

⁴⁴We prefer to adjust the outcomes for Wallonia because we consider it as the control region.

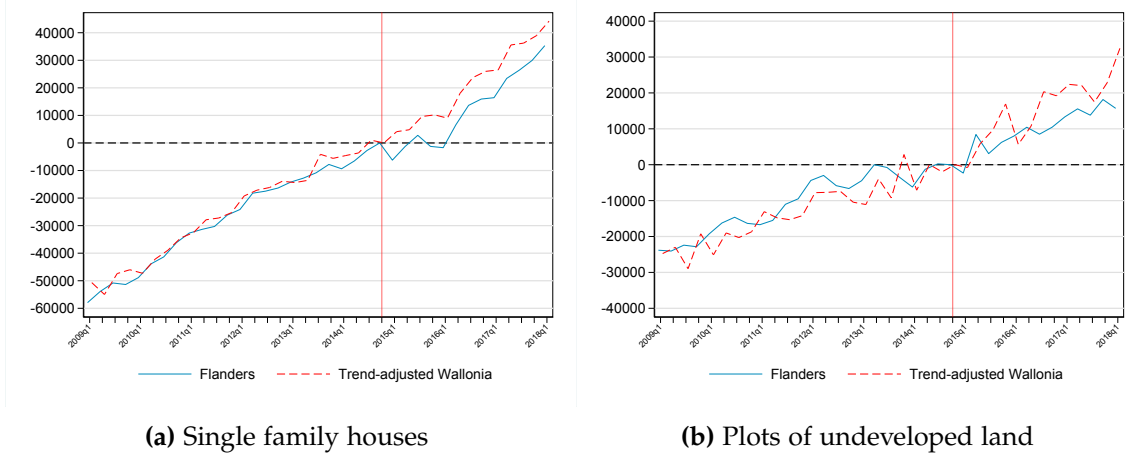


Fig 7. The average change in quality-adjusted prices accounting for the difference over the pre-reform period

Notes: This figure plots the estimates of δ_t in Equation 12 for single family houses in panel (a) and the estimates of δ_t in Equation 11 for plots of undeveloped land in panel (b). The vertical line indicates the reference period 2014Q4, the last quarter before the start of the reforms. Panel (a) in Figure B.1 in the Online Appendix plots the estimates of δ_t in Equation 11 for single family houses..

Sources: sales data from GAPD, authors' calculations.

The last three assumptions concern treatment effect heterogeneity. These assumptions are: i) no treatment effect dynamics, ii) homogeneous average causal responses (ACR) across groups and iii) homogeneous average causal responses across doses. Callaway et al. (2021) define the average causal response of dose d as: $ACR(d) = \frac{\partial E[Y_t(d)]}{\partial d}$. In the case of prices, this estimand has a familiar interpretation as the capitalization rate of the housing subsidies. Together, the three assumptions imply that changes in the present value of housing subsidies immediately capitalize in prices to the same degree in Flanders and Wallonia. We need to make these assumptions because both regions are treated before and after the reforms and, small changes notwithstanding, the average doses change by a different amount two times over the sample period, i.e. in 2015 and in 2016. The estimate of β will include nuisance terms in addition to the true treatment effect if one or more of these assumptions is violated.

6 Results

This section presents the estimates of the effect of a change in the present value of future tax subsidies on the equilibrium prices and quantities of houses and land. The results confirm the theoretical predictions in Section 2. Changes in the present value of future subsidies largely capitalize in the price of houses and plots of undeveloped land while their equilibrium quantities remain unaffected. We first present the estimates of the effect on prices after which we discuss the estimates of the effect on quantities.

6.1 Prices

Before we present the estimates of the average treatment effect, we first more formally assess the parallel trends assumption by estimating the following specification:

$$\Delta \tilde{P}_{r,t} = \beta_t \times \text{Flanders}_r + \alpha_r \times \text{Flanders}_r + \delta_t + \varepsilon_{i,r,t} \quad (13)$$

and plotting the estimates of β_t .⁴⁵ The estimates in Figure 8 confirm the two key take-aways from the plots in Figure 7. First, the average change in quality-adjusted prices in Flanders and the trend-adjusted average change in quality-adjusted prices in Wallonia are closely in line over the pre-reform period. The difference between both series fluctuates around zero. Second, quality-adjusted prices in Flanders clearly decreased relative to trend- and quality-adjusted prices in Wallonia after 2014.

⁴⁵Under the assumptions discussed in Section 5, the estimates of β_t can be interpreted as quarterly average treatment effects. We are, however, not able to compute standard errors since there are only two units in the sample.

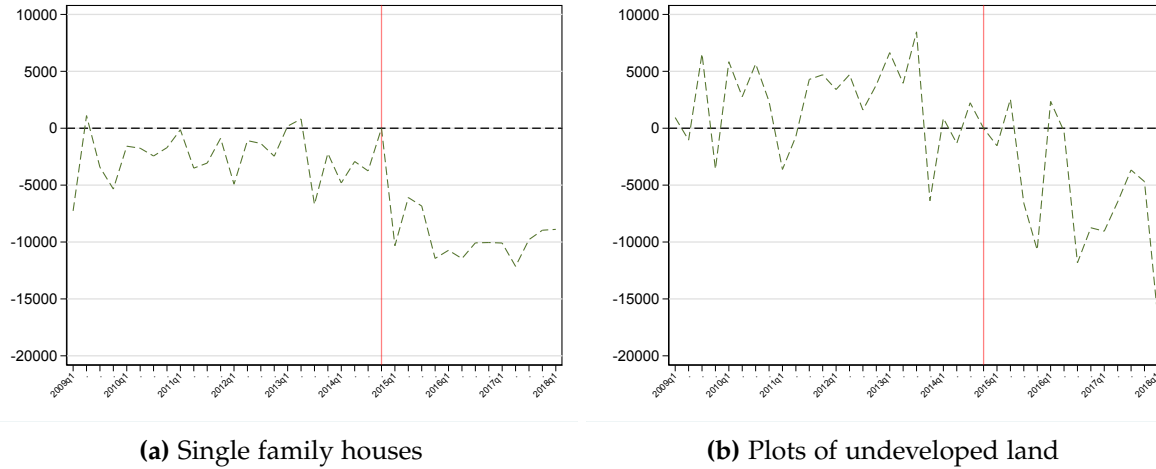


Fig 8. The difference in the average change in quality-adjusted prices

Notes: This figure plots the estimates of β_t in Equation 13 using the average change in quality-adjusted prices of single family houses (panel a) and plots of undeveloped land (panel b) as dependent variable. The average change in quality-adjusted house prices are estimates of δ_t from the repeat sales model in Equation 12, while the average change in quality-adjusted land prices are the estimates of δ_t from the hedonic pricing model in Equation 11. The average changes in quality-adjusted prices in Wallonia are adjusted for the difference in pre-reform trends by subtracting the estimate of the region-specific linear trend. The (red) vertical line indicates 2014Q4, the reference period and last quarter before the reforms of the MPD started in both regions. Figure B.2 in the Online Appendix plots these estimates together with the estimates of β_t in Equation 13 using the average change in quality-adjusted prices without the adjustment for differential pre-trends as dependent variable.

Sources: sales data from GAPD, authors' calculations.

The results in Table 2 show that changes in the average present value of future subsidies largely capitalize in house and land prices. The first row presents the estimates of β in Equation 9 which can be interpreted as the ATE(d) under the assumptions discussed in Section 5. The estimate in column one indicates that quality-adjusted prices of single family houses in Flanders dropped, on average, 7,293 euro relative to quality-adjusted prices of single family houses in Wallonia. The estimate in column four shows that quality-adjusted prices of undeveloped plots in Flanders decreased, on average, 7,969 euro relative to quality-adjusted prices of undeveloped plots in Wallonia. The third row in Table 2 shows the estimates of β divided by d , i.e. 7,286.26 euro. These numbers can be interpreted as the ACR(d) if we assume the ACR is homogeneous across doses.⁴⁶ The

⁴⁶Alternatively, we could replace the interaction in Equation 9 by the regional average PVs we computed in Section 3. However, these average PVs might differ from the true regional average PVs. Measurement error in the dose will bias the estimates of β towards zero.

estimate in the first column indicates that the change in the PV of future subsidies completely capitalized in house prices. The rate at which the change in the PV capitalized in land prices is almost ten percentage points higher. We cannot, however, reject the hypothesis that it is equal to one.

The estimates in columns one and four might be downward biased if households moved from Flanders to Wallonia to benefit from the more generous housing subsidy. The relatively larger reduction in the PV would then not only decreased demand in Flanders but also increased demand in Wallonia.⁴⁷ To assess whether the estimates are biased due to household sorting, we exclude sales within 5 and 10 kilometre of the regional border and re-run the complete analysis including the quality- and trend-adjustments. Columns two and three show that the estimate of the effect on house prices does not increase when we drop sales close to the regional border. In contrast, the estimates of the effect on land prices in columns five and six are 25% higher than the estimate in column four. Given that there is no sign the latter two estimates might be biased in a different way, we believe these are more credible estimates of the true treatment effect.⁴⁸

We can think of two potential, not mutually exclusive, drivers of the difference in the effect on house and land prices. First, the supply of land might be more price-elastic than the supply of existing houses.⁴⁹ A similar change in the present value of future tax subsidies will then capitalize to a smaller degree in land prices. The second reason is that the present value of the subsidies for the Flemish households who bought the houses in the repeat sales sample might have decreased more relative to their Walloon counterparts than the present value of future subsidies for the Flemish households who bought the plots of undeveloped land. The Flemish Housing Survey of 2013 shows that

⁴⁷This would be a violation of the stable unit treatment value (SUTVA) assumption (Imbens and Wooldridge, 2009).

⁴⁸Figure B.3 in the Online Appendix plots the estimates of β_t in Equation 13 for the 'donut' samples.

⁴⁹In addition to this difference in the price-elasticities of supply, changes in the average present value would also capitalize to a smaller degree in land prices than house prices if the demand for land is less price-elastic than the demand for existing houses. However, we think this is unlikely.

the monthly disposable income of households who bought an existing house between 2009 and 2013 is 20% less than the income of households who bought or built a new house.⁵⁰ The average present value of subsidies for a Flemish household in the lower half of the taxable income distribution decreased 9,352 euro relative to the value for their Walloon counterparts while it decreased only 4,972 euro for a Flemish household in the upper half of the distribution.

Lastly, Table B.4 in the Online Appendix reports three additional sets of estimates based on average changes in quality-adjusted prices which we estimated using the following three different methods: i) the hedonic pricing method using the full sample of house sales, ii) the repeat sales method including the observed characteristics and iii) the hedonic pricing method using the repeat sales sample. These additional estimates of the capitalization rate are both qualitatively as well as quantitatively similar to the estimates presented in Table B.4. Their size ranges from 0.86 (0.07) to 1.09 (0.11).

⁵⁰Specifically, households who bought an existing house report an average monthly disposable income of 2,812 euro, while households who bought or built a new house report an average monthly income of 3,363 euro. The 550 euro difference is significantly different from zero.

Table 2. The average effect of housing subsidies on house and land prices

	Price of single family houses ($P_{h,t}$)			Price of undeveloped land ($P_{l,t}$)		
	Full sample (1)	> 5km (2)	> 10km (3)	Full sample (4)	> 5km (5)	> 10km (6)
ATE(d)	-7,293*** (661.4)	-7,637*** (836.2)	-7,471*** (891.4)	-7,969*** (1,735)	-5,820*** (1,737)	-5,698*** (1,945)
ACR(d)	1.00*** (0.09)	1.05*** (0.12)	1.03*** (0.12)	1.09*** (0.24)	0.80*** (0.24)	0.78*** (0.27)
Obs.	74	74	74	74	74	74
Sales	99,069	90,087	81,751	34,706	29,168	27,305
Plots	47,671	43,399	39,388	32,082	26,851	25,151

Notes: The first row in this table presents estimates of β from the two-way fixed effects model in Equation 9 using the trend-adjusted average change in quality-adjusted prices of houses and plots of land as the dependent variable ($\Delta \hat{P}_{r,t}$). The average change in quality-adjusted house prices are estimates of δ_t from the repeat sales model in Equation 12, while the average change in quality-adjusted land prices are the estimates of δ_t from the hedonic pricing model in Equation 11. The average changes in quality-adjusted prices in Wallonia are adjusted for the difference in pre-reform trends by subtracting the estimate of region-specific linear trend. The third row reports the estimates of β divided by the dose d as computed in Equation 10, i.e. 7,286.26 euro. Robust standard errors are reported in parentheses. See Table B.4 for the same set of estimates based on the average change in house prices adjusted for quality using the hedonic pricing method on the full and repeat sales sample.

Sources: sales data from GAPD, authors' calculations.

6.2 Quantities

Figure 9 plots the estimates of β_t in Equation 13 using the trend-adjusted logarithm of sales of single family houses (panel a) and plots of undeveloped land (panel b) as dependent variable. There are three takeaways. First, the trends in these series were in line over the pre-reform period since the estimates fluctuate around zero. Second, there is a clear peak in sales in 2014Q4 and a trough during the following quarters. These estimates suggests that relatively more households anticipated the reforms in Flanders than in Wallonia. Recall that we address the potential bias due to intertemporal substitution by excluding 2014Q4 to 2016Q1 in the analyses. Third, sales in Flanders does not seem to have decreased relative to Wallonia after 2014. Comparing the estimates in Figure 9 to the estimates β_t using the unadjusted logarithm of sales suggests that the large differ-

ence in the average quarterly number of sales we pointed out in the data section is the result of a difference in pre-reform trends (see Figure C.1 in the Online Appendix).

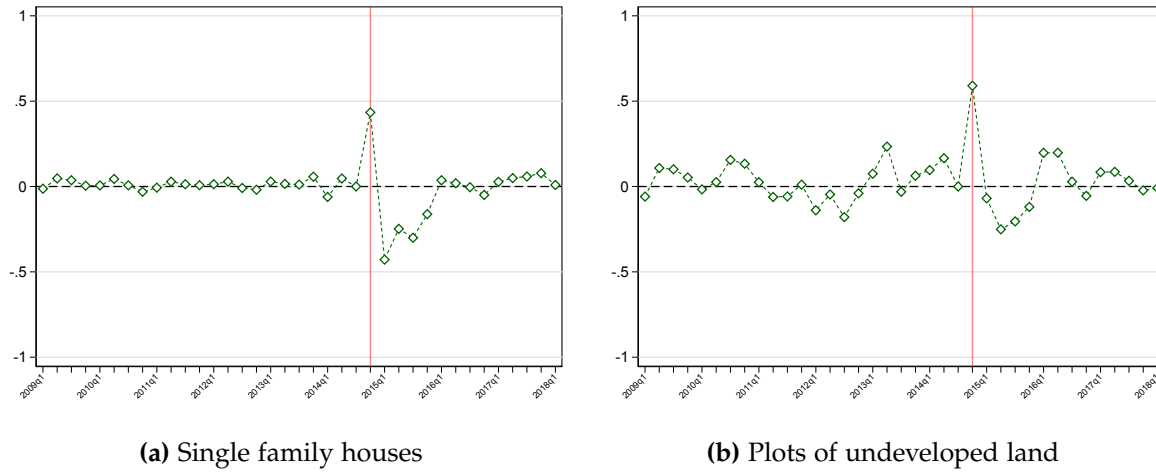


Fig 9. The difference in the trends in sales

Notes: This figure plots the estimates of β_t in Equation 13 using the trend-adjusted logarithm of sales of single family houses (panel a) and plots of undeveloped land (panel b) as dependent variable. The baseline quarter is 2014Q3. The (red) vertical line indicates 2014Q4, the last quarter before the reforms of the MPD. See Figure C.1 in the Online Appendix for the estimates of β_t using the unadjusted logarithm of sales as dependent variable.

Sources: sales data from GADP, authors' calculations.

The first three columns of Table 3 present the estimates of the effect on sales of existing houses. In the theoretical framework in Section 2, we assumed a fraction of the stock of existing houses is supplied inelastically to the market and the market clears. Both assumptions might not hold in practice however. First, the reduction in the housing tax subsidies might not only have reduced demand of first-time buyers but might also have withheld current owners of moving or trading-up, which would lower the supply of existing houses. Second, a number of houses always remain vacant, and the vacancy rate might have increased due to the decrease in demand for houses. Despite these additional margins of response, the estimate in the first column indicates that sales of existing houses in Flanders did not significantly decrease relative to sales of existing houses in Wallonia.

The effect on the full sample of sales might hide a shift from buy-to-occupy (BTO)

to buy-to-let (BTL) sales. Investors benefited from the decline in prices, while their tax reduction was not affected by the reforms.⁵¹ We split sales of existing houses in BTO and BTL sales based on whether or not a rental contract is registered within 180 days after the sale.⁵² The estimates in columns two and three are both not significantly different from zero. The lack of a shift from BTO to BTL sales can be the result of the reduction in the transaction costs and down payment due to the decrease in house prices. For credit-constrained households in the lower tax brackets, the limited decrease in the present value of future subsidies might be outweighed by the reduction in the transactions costs, making it optimal for them to become homeowners (Sommer and Sullivan, 2018).

The estimate in column four in Table 3 indicates that the equilibrium quantity of undeveloped plots in Flanders did not significantly decrease relative to Wallonia after the reforms. This does not change when we restrict the sample of undeveloped plots to plots which are described as land for residential development by both the deed and the land register.⁵³ The estimates of the effect on sales in columns one and four suggest that the difference in the effect on house and land prices in the previous section is likely due to a difference in average doses rather than a difference in the price-elasticities of supply. Sales of undeveloped plots should have decreased more than sales of existing houses if its supply is more price-elastic.

As we pointed out in the data section, developers might have kept on buying land while they reduced construction. Figure 10 plots the estimates of β_t in Equation 13 using the trend-adjusted logarithm of construction permits for single family houses as dependent variable. The estimates also fluctuate around zero over the pre-reform period

⁵¹Households who owned more than one house were not eligible for the MPD but enjoyed a different smaller tax reduction. See Goeyvaerts and Buyst (2019) for more information on the user cost of rental housing in Belgium.

⁵²We matched the sales of single family houses to the universe of rental contracts in Belgium.

⁵³This estimate is, however, potentially upward biased due to the change in the way the GADP registered sales of split plots in 2015. Although this change affected both regions, Table C.1 in the Online Appendix shows that the share of plots that are also described as land for residential development by the land register increased 31 percentage points in Flanders, while it only increased 22 percentage points in Wallonia.

and do not decrease after 2014. The latter is confirmed by the estimate in the last column of Table 3, which is positive but not significantly different from zero.

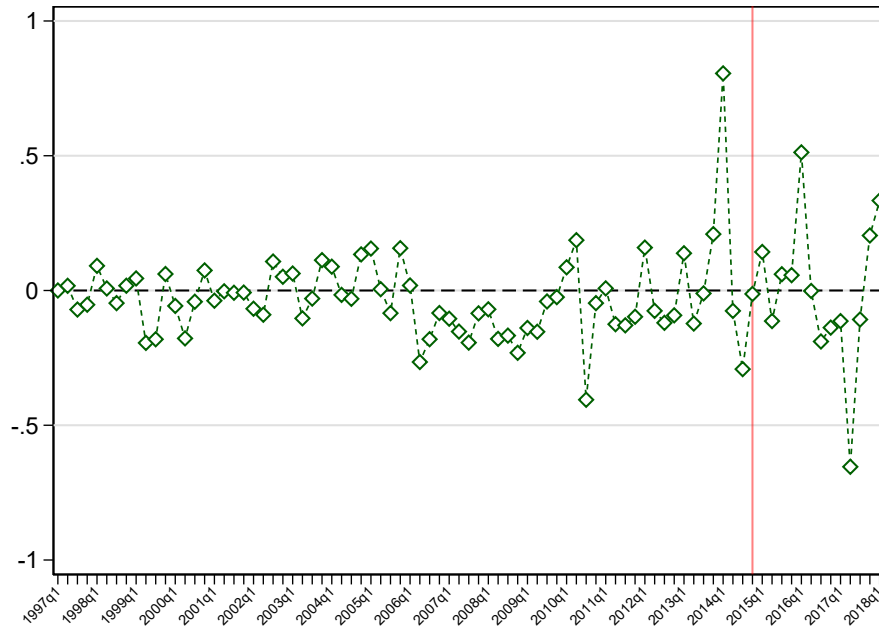


Fig 10. The difference in the trends in permits

Notes: This figure plots the estimates of β_t in Equation 13 using the trend-adjusted logarithm of construction permits for single family houses as dependent variable. The (red) vertical line indicates 2014Q4 which is the baseline quarter and the last quarter before the reforms of the housing subsidies. See Figure C.2 in the Online Appendix for the estimates of β_t using the unadjusted logarithm of construction permits as dependent variable.

Sources: permit data from Statbel, authors' calculations.

Table 3. The effect of housing subsidies on the quantity of housing and land

	Sales of existing houses ($Q_{h,t}^E$)			Undeveloped plots ($Q_{l,t}^*$)		Permits ($Q_{h,t}^N$)
	Full	BTO	BTL	Full	Restricted	
	(1)	(2)	(3)	(4)	(5)	(6)
ATE(d)	0.0188	0.0211	0.0143	0.0025	0.0536	0.0289
	(0.0150)	(0.0151)	(0.0604)	(0.0346)	(0.0545)	(0.0785)

Notes: This table reports the estimates of β from the two-way fixed effect model in Equation 9 using the logarithm of sales and construction permits as dependent variables. Sales of existing houses is split into buy-to-occupy (BTO) and buy-to-let (BTL) sales in columns two and three. Column four reports the estimate of the effect on sales of undeveloped plots for residential development. In column five, we restrict the sample to sales which are described as land for residential development by both the deed and the register. We drop 2014Q1 to 2016Q1 for the sales analyses resulting in a sample of 60 observations. The number of observations is 170 for permits. Robust standard errors in parentheses.

Sources: sales data from GAPD and permit data from Statbel, authors' calculations.

7 Conclusion

In this paper, we studied the incidence of tax subsidies for homeowners. Reforms of the Belgian mortgage payment deduction (MPD) created variation across regions and over time which we exploited to credibly identify the effect on the prices and quantities of housing and land. We reported two major findings. First, the average present value of future tax subsidies largely capitalizes into house and land prices. According to our preferred specifications, the capitalization rate is 100% for single family houses and 80% for land. In both cases, we cannot reject the null hypothesis of full capitalization. Second, the much larger reduction in the average value of the subsidy in Flanders did not decrease sales of existing houses, sales of plots of undeveloped land nor construction permits for single-family houses relative to Wallonia.

The results in this paper show that the tax subsidies largely accrued to current property owners instead of future home buyers. However, the economic incidence of subsidies depends on the relative size of the price-elasticities of demand and supply, which can vary along different dimensions. The literature studying the incidence of non-recurrent tax subsidies for homeowners provides an interesting example. [Berger et al. \(2020\)](#) finds that the First-Time Homebuyer Credit (FTHC) increased sales in the U.S. by almost 10% with only a small increase in prices. In contrast, [Krolage \(2023\)](#) shows that a subsidy of similar size completely capitalized in listing prices in Bavaria. Importantly, the FTHC was introduced during the aftermath of the Great Recession. This was a time when housing demand was low as vacancies were on the rise and inventory was at historic levels.

Due to data limitations, we were not able to examine heterogeneity along a couple of interesting dimensions. Future studies might be able to take into account. First, the relative size of the price-elasticities can vary over space. [Hilber and Turner \(2014\)](#) show that the mortgage subsidy rate is negatively correlated with homeownership and positively correlated with house prices in tightly regulated metropolitan areas, whereas it is

positively correlated with homeownership and not significantly correlated with prices in lightly regulated areas.⁵⁴ These results suggest that future home buyers benefit more from housing subsidies in areas where housing supply is more responsive. However, [Baum-Snow and Han \(2022\)](#) show that the price-elasticity of supply can vary significantly even within metropolitan areas. Future research could exploit increasingly granular data to study how the incidence varies with local supply conditions.

Second, the relative size of the elasticities might vary over the income distribution. [Davis \(2019\)](#) shows that the effect of a one percentage point rise in the mortgage interest rate subsidy increases monotonically in house quality. Assuming house quality is strongly correlated with household income, these results suggest that buyers in the lower part of the income distribution benefit more from housing subsidies than buyers in the upper end of the distribution. Future research may be able to link house sale records to information on the socio-economic characteristics of the buyers to investigate heterogeneity in this dimension in more detail.

Third, the relative size of the price-elasticities might vary over time. This paper finds that changes in the discounted value of future tax subsidies largely capitalize in prices over a period up to three years after the reforms. This period of time can be considered as the short-run. If the price-elasticity of supply of housing increases relative to the price-elasticity of demand over time, the rate at which the subsidies capitalize into house prices might decrease while the equilibrium quantity of housing increases. While challenging, future studies could try to examine the incidence over different periods in time.

⁵⁴See also [Carozzi et al. \(2020\)](#) who find that the Help-to-Buy program in the U.K. raised prices in the Greater London Area without a significant effect on construction, whereas it raised construction without a significant effect on prices at the border with Wales.

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ONLINE APPENDIX NOT FOR PUBLICATION

A Additional information, tables and figures on the institutional background

A.1 The average deducted mortgage payment and the marginal tax rate

Statistics Belgium provided us the total number of tax payers that deduct a mortgage payment, the total amount that is deducted and the total loss in revenue per tax bracket from 2005 to 2013 for each Belgian region. We use these data to compute the average deducted mortgage payment and the average marginal tax rate at which the payment is deducted. These are reported in Panel B and C of Table A.4. The first row in Panel B reports the maximum amount including the add-on for the first ten years, while row two to four report the weighted average deducted mortgage payment for Belgium, Flanders and Wallonia respectively.⁵⁵ The weights are the shares of each tax bracket in the total number of tax payers that deduct a mortgage payment, which are reported in Table A.8. Row five reports the difference in the average deducted mortgage payment between Flanders and Wallonia. Right after the introduction of the MPD, in 2005, the average deducted mortgage payment is almost 1,000 euro less than the maximum deductible. It increases to just about 300 euro below the cap after three years.

Despite that the average deducted mortgage payment is less than the maximum amount, we assume that the cap was binding for most home buyers. First, the amount a tax payer had to borrow in order to deduct the maximum is significantly lower than the average sales price at that time. For example, at the average mortgage interest rate in 2014, a couple only had to borrow 88,000 euro for both of them to be able to deduct the maximum of 3,040 euro. This is considerably lower than the average sales price

⁵⁵We compare the average mortgage payment to the maximum amount including the add-on for the first ten years because most if not all tax payers that deducted a mortgage payment will have a mortgage less than ten years old at that time.

at that time, which was 239,233.25 euro. Furthermore, the third (2017) wave of the European Household Finance and Consumption Survey (HFCS) shows that the average annual mortgage payment for mortgages with the main purpose to buy a house is 10,176 euro.⁵⁶ Second, Table A.6 shows that the average deducted mortgage payment does not vary substantially across tax brackets. The difference between the lowest average deducted mortgage payment in the 30% bracket and the highest average payment in the 50% bracket is about 170 euro.

We can think of two reasons why the average deducted mortgage payment is smaller than the maximum amount. First, tax payers do not make a 'full' mortgage payment in the first year of their mortgage. The size of the payment in the first year of the mortgage depends on the date of the mortgage closing. The first payment for mortgages closed at the end of the year might be considerably smaller than the cap. This explains the relative large difference during the first years after the introduction of the MPD. However, its impact on the average deducted mortgage payment declines as the share of tax payers that deduct a full annual mortgage payment grows over time.

Second, tax payers could not only deduct the payment for a mortgage to buy or build a house, but also payments for a mortgage to renovate a house.⁵⁷ Aggregate data from the Belgian Association of Creditors shows that 31% of all mortgages closed from 2005 to 2013 were used to renovate a house.⁵⁸ The average amount households borrowed to renovate was 33,850 euro. At the average interest rate during this period, the annual payment for a 20-year mortgage was 2,603 euro and 4,231 euro for a 10-year mortgage. These annual payments would not allow a couple to deduct the maximum amount each.

Panel C in Table A.4 provides the weighted average rate at which the mortgage pay-

⁵⁶The respondents in the HFCS provide the monthly mortgage payment, which is on average 848 euro. We excluded refinances.

⁵⁷In addition to buying, building and renovating a house, households could also deduct payments for a mortgage to 'retain' a house, e.g. a mortgage to pay the estate tax if they inherited a house.

⁵⁸The association splits the aggregate data in seven categories based on the purpose of the mortgage: 1) purchase, 2) construction, 3) renovation, 4) purchase with renovation, 5) other, 6) refinances (internal+external) and 7) refinances (external).

ment is deducted. The average marginal tax rate of tax payers who deduct a mortgage payment fluctuates around 42.1%. The average marginal income tax rate does not, however, account for the flat municipal surcharge on the personal income tax, which was 7.4% over this period. The total average marginal tax rate was therefore 45.2%.

Table A.4. The average deducted mortgage payment and the average marginal tax rate

	2005	2006	2007	2008	2009	2010	2011	2012	2013
A. The number of tax payers that deduct mortgage payments (thousands)									
Belgium	150.6	320.6	483.4	643.0	808.9	972.2	1134.8	1269.1	1381.3
Flanders	97.2	207.8	311.8	416.3	528.4	638.1	747.4	835.7	910.0
Wallonia	43.4	92.1	140.6	186.6	232.6	278.7	324.1	364.1	396.9
B. The cap and the weighted average mortgage payment (euro)									
Cap	2,490	2,560	2,600	2,650	2,770	2,770	2,830	2,930	3,010
Belgium	1,567.2	2,065.6	2,241.5	2,353.2	2,440.8	2,478.1	2,539.8	2,664.2	2,740.3
Flanders	1,521.8	2,021.0	2,203.3	2,339.6	2,433.6	2,462.4	2,522.1	2,633.9	2,712.0
Wallonia	1,443.4	1,973.3	2,178.0	2,330.1	2,446.5	2,476.7	2,544.9	2,654.6	2,736.9
Difference	78.4	47.7	25.3	9.5	-12.9	-14.3	-22.9	-20.7	-24.9
C. The weighted average marginal tax rate (%)									
Belgium	41.9	41.7	41.7	42.0	41.8	42.0	42.1	42.2	42.2
Flanders	42.4	42.2	42.2	42.5	42.3	42.4	42.6	42.6	42.6
Wallonia	41.1	40.8	40.8	41.2	40.9	41.2	41.4	41.4	41.4

Notes: Panel A reports the total number of tax payers that deduct mortgage payments in Belgium, Flanders and Wallonia. Panel B and C report the average deducted mortgage payment and the average marginal tax at which this mortgage payment is deducted.

Sources: Income tax data from Statistics Belgium, authors' calculations.

Table A.5. Income thresholds of the tax brackets in the personal income tax (euro)

Tax bracket		I 25%	II 30%	III 40%	IV 45%	V 50%
2005	5,780	7,100	10,100	16,830	30,840	
2006	5,940	7,290	10,380	17,300	31,700	
2007	6,040	7,420	10,570	17,610	32,270	
2008	6,150	7,560	10,760	17,920	32,860	
2009	6,690	7,900	11,240	18,730	34,330	
2010	6,690	7,900	11,240	18,730	34,330	
2011	6,830	8,070	11,480	19,130	35,060	
2012	6,800	8,350	11,890	19,810	36,300	
2013	6,990	8,590	12,220	20,370	37,330	
2014	7,070	8,680	12,360	20,600	37,750	
2015	7,090	8,710	12,400	20,660	37,870	
2016	7,130	10,860	12,470	20,780	38,080	
2017	7,270	11,070	12,720	21,190	38,930	
2018	7,430	12,990		22,290	39,660	

Notes: This table reports the income thresholds of the five tax brackets in the personal income tax.

Table A.6. The average deducted mortgage payment per tax bracket (euro)

Tax bracket	I 25% (1)	II 30% (2)	III 40% (3)	VI 45% (4)	V 50% (5)	Average (6)
A. Belgium						
2005	1,611.60	1,283.54	1,527.04	1,498.51	1,567.19	1,509.34
2006	2,080.66	1,792.93	2,021.83	2,009.71	2,065.62	2,012.84
2007	2,251.28	1,987.73	2,200.70	2,206.64	2,241.46	2,200.72
2008	2,388.05	2,192.10	2,344.06	2,344.72	2,353.19	2,340.76
2009	2,499.70	2,308.40	2,451.58	2,447.87	2,440.84	2,443.44
2010	2,527.35	2,326.03	2,470.65	2,478.16	2,478.09	2,471.30
2011	2,585.38	2,389.78	2,528.57	2,544.14	2,539.80	2,533.80
2012	2,686.39	2,489.67	2,624.65	2,658.83	2,664.17	2,644.34
2013	2,765.17	2,571.48	2,703.75	2,739.58	2,740.28	2,723.65
B. Flanders						
2005	1,647.31	1,261.89	1,552.99	1,511.18	1,561.09	1,521.79
2006	2,114.88	1,769.44	2,034.40	2,019.96	2,054.14	2,021.03
2007	2,270.95	1,947.71	2,204.87	2,211.36	2,233.37	2,203.29
2008	2,405.34	2,156.82	2,347.11	2,344.38	2,343.44	2,339.58
2009	2,514.37	2,262.38	2,446.84	2,439.91	2,418.49	2,433.55
2010	2,536.83	2,275.16	2,461.93	2,471.41	2,465.89	2,462.43
2011	2,589.37	2,333.29	2,514.73	2,536.03	2,525.34	2,522.06
2012	2,689.25	2,429.08	2,608.38	2,651.46	2,656.49	2,633.90
2013	2,767.69	2,515.48	2,685.68	2,730.60	2,731.42	2,712.03
C. Wallonia						
2005	1,485.23	1,269.62	1,454.30	1,443.70	1,509.16	1,443.42
2006	2,002.50	1,789.94	1,989.12	1,970.62	2,048.62	1,973.29
2007	2,199.43	2,010.99	2,180.80	2,184.64	2,235.83	2,177.95
2008	2,342.12	2,211.96	2,331.25	2,338.97	2,353.33	2,330.08
2009	2,450.99	2,342.98	2,448.18	2,455.94	2,468.65	2,446.48
2010	2,491.18	2,369.33	2,477.66	2,487.98	2,487.50	2,476.71
2011	2,557.93	2,445.20	2,544.02	2,555.78	2,555.68	2,544.93
2012	2,663.82	2,553.92	2,644.20	2,670.67	2,666.44	2,654.56
2013	2,741.62	2,630.13	2,727.05	2,755.41	2,747.54	2,736.89

Notes: This table reports the average deducted mortgage payment per tax bracket.

Sources: Income tax data from Statistics Belgium, authors' calculations.

Table A.7. The distribution of tax payers (%)

Tax bracket	I 25% (1)	II 30% (2)	III 40% (3)	VI 45% (4)	V 50% (5)
A. Belgium					
2005	16.8	14.3	25.4	26.0	10.9
2006	17.3	14.2	25.1	26.2	10.8
2007	17.2	13.4	25.5	26.3	11.0
2008	16.7	12.0	26.0	26.9	11.6
2009	24.2	11.8	26.5	26.7	10.8
2010	24.4	11.1	26.8	26.7	11.0
2011	23.0	10.8	27.4	27.4	11.4
2012	22.1	10.7	27.8	27.9	11.5
2013	22.3	10.6	28.0	27.9	11.3
Average	22.9	10.9	27.3	27.5	11.4
B. Flanders					
2005	16.6	13.8	25.2	27.7	11.7
2006	17.0	13.7	25.0	27.9	11.6
2007	16.9	12.9	25.4	28.1	11.8
2008	16.4	11.5	25.8	28.7	12.5
2009	22.1	11.3	26.6	28.5	11.6
2010	22.0	10.6	27.0	28.5	11.9
2011	20.6	10.3	27.5	29.3	12.3
2012	19.6	10.2	28.0	29.8	12.4
2013	19.6	10.0	28.3	29.8	12.2
Average	7.5	5.8	19.5	47.4	19.8
C. Wallonia					
2005	16.7	14.9	26.2	24.4	9.5
2006	17.2	14.8	25.9	24.8	9.5
2007	17.1	14.1	26.3	25.0	9.8
2008	16.6	12.5	26.9	25.7	10.4
2009	25.1	12.4	27.2	25.5	9.8
2010	25.1	11.7	27.5	25.7	10.0
2011	23.8	11.3	28.0	26.5	10.4
2012	22.9	11.2	28.4	27.0	10.5
2013	23.2	11.1	28.5	26.9	10.4
Average	23.8	11.5	28.0	26.5	10.3

Notes: This table reports the share of tax payers that file a return per tax bracket.

Table A.8. The distribution of tax payers that deduct a mortgage payment (%)

Tax bracket	I 25% (1)	II 30% (2)	III 40% (3)	VI 45% (4)	V 50% (5)	$\bar{\tau}$ (6)
A. Belgium						
2005	7.9	7.1	21.0	47.6	16.3	41.9
2006	9.2	6.8	21.2	47.1	15.7	41.7
2007	9.5	6.7	21.0	46.4	16.4	41.7
2008	9.1	6.1	19.9	46.8	18.1	42.0
2009	9.5	6.4	20.9	46.0	17.2	41.8
2010	9.1	6.4	20.4	45.8	18.1	42.0
2011	8.7	6.5	20.0	45.7	19.1	42.1
2012	8.6	6.5	19.9	45.6	19.4	42.2
2013	8.5	6.6	20.1	45.6	19.3	42.2
Average	8.7	6.5	20.2	45.7	18.9	42.1
B. Flanders						
2005	6,8	6,2	20,1	50,1	16,8	42,4
2006	7.9	6.0	20.3	49.5	16.3	42.2
2007	8.1	5.8	20.0	48.9	17.2	42.2
2008	7.8	5.3	19.0	49.0	19.0	42.5
2009	8.1	5.6	20.1	48.1	18.1	42.3
2010	7.9	5.7	19.7	47.7	19.0	42.4
2011	7.5	5.8	19.3	47.3	20.1	42.6
2012	7.4	5.9	19.2	47.1	20.4	42.6
2013	7.3	5.9	19.5	47.1	20.2	42.6
Average	7.5	5.8	19.5	47.4	19.8	42.5
C. Wallonia						
2005	9.3	9.0	23.5	44.5	13.7	41.1
2006	11.1	8.4	23.6	44.0	12.8	40.8
2007	11.4	8.3	23.4	43.4	13.5	40.8
2008	10.8	7.6	22.0	44.5	15.0	41.2
2009	11.4	7.9	23.0	43.5	14.2	40.9
2010	10.9	7.8	22.3	43.9	15.1	41.2
2011	10.3	7.7	21.6	44.3	16.1	41.4
2012	10.1	7.7	21.6	44.2	16.3	41.4
2013	10.0	7.8	21.6	44.2	16.4	41.4
Average	10.4	7.8	21.9	44.1	15.9	41.3

Notes: Columns 1 to 5 in this table report the share of tax payers that deduct a mortgage payment per tax bracket. Column 6 reports the weighted average marginal tax rate at which the mortgage payment is deducted, which is computed using these shares as weights.

Sources: Income tax data from Statistics Belgium, authors' calculations.

A.2 The present value of future tax subsidies

We start with the present value of the federal mortgage payment deduction for the period of 2005 to 2014. The PV of future MPD's for an individual tax payer that buys a home in region r during period t and whose marginal euro of personal income falls in tax bracket b , can be computed as follows:

$$PV_{r,b,t} = \sum_{n=1}^{10} (1+r_t)^{-n} \times \tau_b \times (1+s_{r,t}) \times MP_{n \leq 10,t}^* + \sum_{n=10}^N (1+r_t)^{-n} \times \tau_b \times (1+s_{r,t}) \times MP_{n > 10,t}^* \quad (14)$$

The first term is the present value of the MPD's over the first ten years of the mortgage. We assume home buyers deduct the maximum deductible amount which is denoted $MP_{n \leq 10,t}^*$. They deduct the maximum at their marginal tax rate τ_b with the average municipal surcharge $s_{r,t}$ on top. Since having a mortgage is a necessary condition to deduct mortgage payments, we use the average mortgage interest rate from the Monetary financial institution Interest Rate (MIR) survey as the discount rate r_t .⁵⁹

The second term is the present value of the MPD's after the first ten years of the mortgage. In line with our assumption regarding the discount rate, we assume home buyers discount future MPD's over their entire mortgage term denoted N . Data from the third wave of the HFCS shows that the average term-to-maturity of mortgages to buy a house was 22.3 years over the period of 2005 to 2017.

The present value of future MPD's for tax payers who bought a house in Flanders after 2014 or in Wallonia in 2015 is computed as in Equation 14, but with the following adjustments: i) the marginal tax rate and the average municipal surcharge are replaced by a fixed rate of 40% and ii) the maximum deductible amount now varies across regions. These adjustments render the following equation:

$$PV_{r,b,t} = \sum_{n=1}^{10} (1+r_t)^{-n} \times 0.4 \times MP_{n \leq 10,r,t}^* + \sum_{n=10}^N (1+r_t)^{-n} \times 0.4 \times MP_{n > 10,r,t}^* \quad (15)$$

⁵⁹The MIR survey is conducted by the National Bank of Belgium.

The present value of future CH's for tax payers who bought a house in Wallonia after 2015 is computed as follows:

$$\begin{aligned} PV_{r,b,t} = & \frac{1,520}{2} \times \left[\sum_{n=1}^{10} (1+r_t)^{-n} + \sum_{n=1}^{20} (1+r_t)^{-n} \right] \\ & - \mathbb{1}[y_{b,t} > 21,000] \times 0.01275 \times \frac{(y_{b,t} - 21,000)}{2} \times \left[\sum_{n=1}^{10} (1+r_t)^{-n} + \sum_{n=1}^{20} (1+r_t)^{-n} \right] \quad (16) \end{aligned}$$

The first term in Equation 16 is the present value of the tax credit of 1,520 euro. Recall that the credit is reduced by half after 10 years and is completely stopped after 20 years. The second term is the present value of the reduction of the tax credit if the taxable income of the tax payer exceeds the threshold of 21,000 euro. In which $y_{b,t}$ denotes the average taxable income of tax payers whose marginal euro of personal income falls in tax bracket b .⁶⁰

Each tax payer can benefit the MPD or CH individually. A couple that files a joint tax return can both deduct the maximum amount if their annual mortgage payment is equal or larger than two times the maximum amount. The HFCS shows that 54% of home buyers had a spouse or partner when the survey was conducted. We multiply the present value for an individual tax payer computed in Equation 14, 15 and 16 by 1.54 to get the present value of future tax subsidies for the average 'set' of home buyers.

Finally, we compute the regional weighted average present value using the shares in Table A.8 as weights:

$$PV_{r,t} = \sum_{b=1}^5 w_{r,b,t} PV_{r,b,t} \quad (17)$$

In which $w_{r,b,t}$ is the share of tax bracket b in the total number of tax payers that deduct a mortgage payment in region r during period t . Since our data on the MPD stops in 2013 and the shares do not strongly vary over time, we use the shares of 2013 as weights in later years. Figure 1 in the main text plots the $PV_{r,t}$ for each region and the difference.

⁶⁰Given that the upper income threshold of the three lower brackets is lower than 21,000 euro, all tax payers in these brackets enjoy the full tax credit of 1,520 euro.

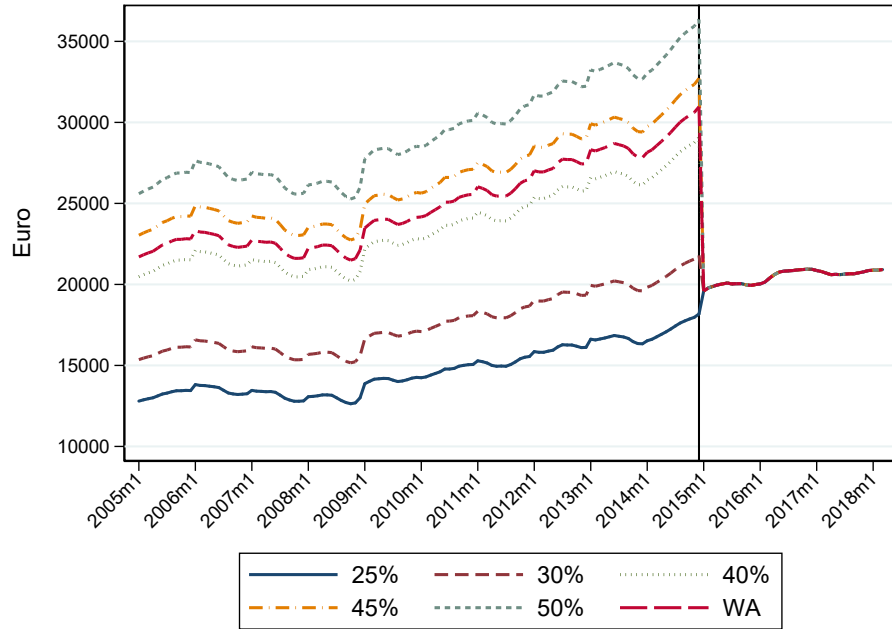
A.3 Variation across tax brackets

The impact of the regional reforms varies across tax brackets. Figure A.11 plots the present value of the future annual tax subsidies per tax bracket ($PV_{r,b,t}$) for Flanders (plot a) and Wallonia (plot b) together with the weighted average present value in each region ($PV_{r,t}$). In Flanders, the present value of the tax subsidies decreased for all tax payers except for those in the lowest tax bracket. In Wallonia, the present value decreased for tax payers in the three upper brackets while it increased for tax payers in the lowest two brackets.

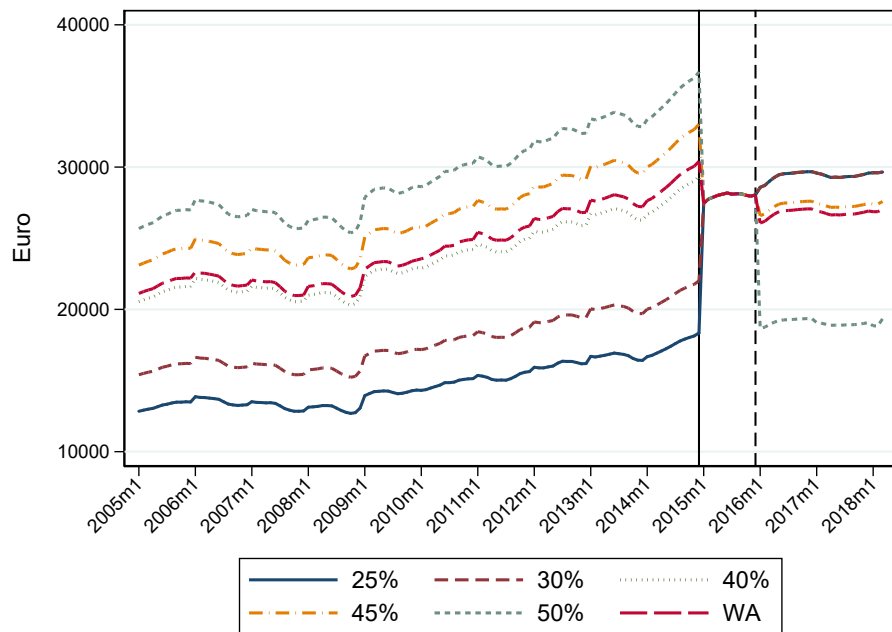
Figure A.12 plots the difference in the present value between Flanders and Wallonia per tax bracket. The difference in the present value between Flanders and Wallonia is quite similar for the four lower tax brackets. The present value of the future subsidies decreases more in Flanders than in Wallonia. The same holds for tax payers in the upper tax bracket in 2015. However, the present value for these tax payers decreases more in Wallonia than in Flanders after the introduction of the Chèque Habitat in 2016.

In the empirical analysis, we exploit variation across the two regions and over time. The plots in Figure A.12 may raise the question whether we can also exploit variation across tax brackets. Gruber et al. (2021), for example, exploit variation across tax brackets to identify the effect of the mortgage interest deduction on homeownership, housing demand and indebtedness in Denmark. Contrary to Gruber and coauthors, however, we do not have any information on the buyers of the houses and plots of land.

An alternative strategy would be to exploit variation across Belgian municipalities. There might be variation in the average present value of future tax subsidies due to variation in the share of tax payers per tax bracket. However, there are a couple of important problems to exploit variation at the municipal-level. First, there are too few sales to properly estimate the average change in quality-adjusted prices for each municipality. Second, this strategy is very vulnerable to spillover effects due to migration. The propensity to migrate is much larger at the municipal level than at the regional level.



(a) Flanders



(b) Wallonia

Fig A.11. The present value of the housing tax subsidies per tax bracket

Notes: This Figure plots the present value of the mortgage payment deduction (MPD) and the Chèque Habitat (CH) per tax bracket ($PV_{r,b,t}$) together with the weighted average present value ($PV_{r,t}$) at a monthly frequency from 2005 to the first quarter of 2018. The solid and dashed vertical lines indicate December 2014 and December 2015 respectively, the last months before new rules from the regional reforms came into force. See Figure A.12 for the difference in the PV.

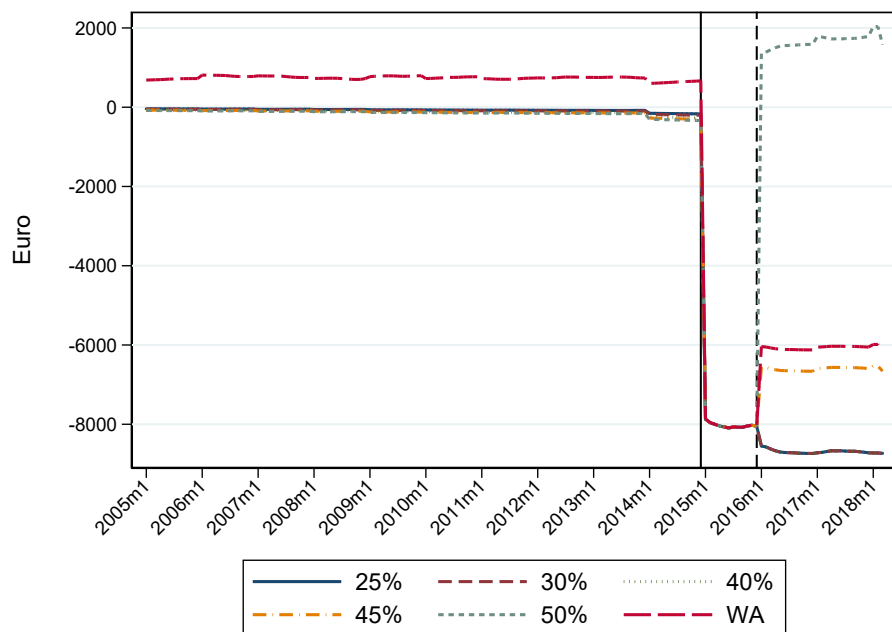


Fig A.12. The difference in the present value of future tax subsidies between Flanders and Wallonia

Notes: This Figure plots the difference in the present value of the MPD between Flanders and Wallonia for each tax bracket.

A.4 The annual tax subsidy

We compute the MPD in the first year of the mortgage as follows:

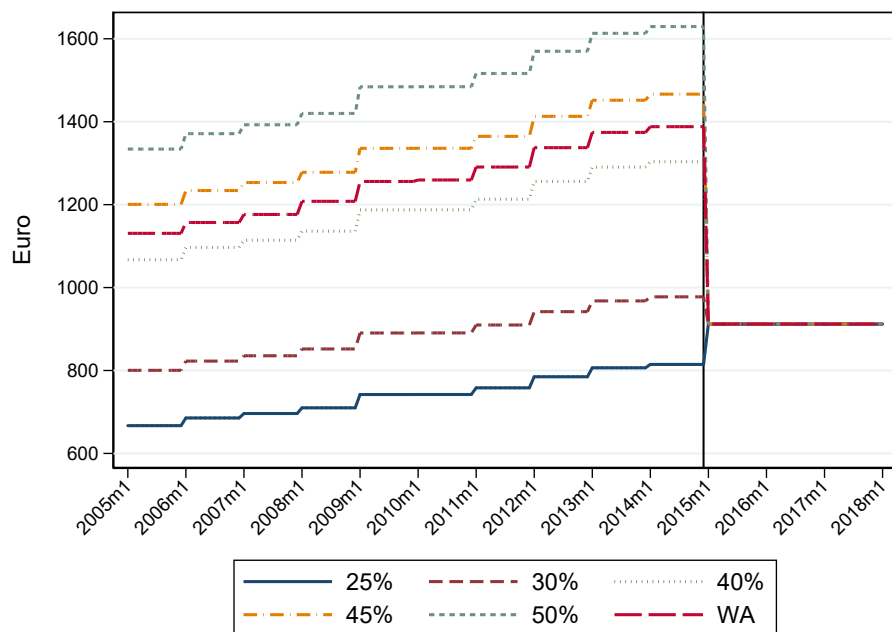
$$\text{MPD}_{r,b,t} = \tau_b \times (1 + s_{r,t}) \times \text{MP}_{n \leq 10, r, t}^* \quad (18)$$

and the Chèque Habitat in the first year as follows:

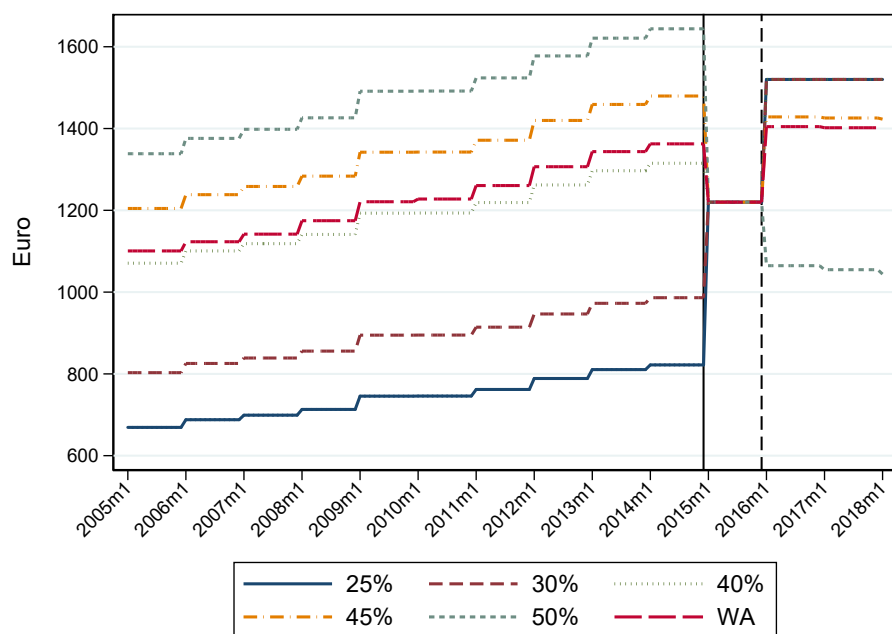
$$\text{CH}_{r,b,t} = 1,520 - 0.01275 \times (\bar{y}_{b,t} - 21,000) \times \mathbb{1}[\bar{y}_{b,t} > 21,000]. \quad (19)$$

Figure A.13 plots the annual tax subsidy per tax bracket for Flanders (plot a) and Wallonia (plot b) together with the weighted average annual subsidy in each region. Panel (b) in Figure A.14 plots the difference in the annual tax subsidies. These plots are very similar to those of the present value of future tax subsidies. The most important difference is that the weighted average annual subsidy in Wallonia increases after the introduction of the Chèque Habitat, while the weighted average present value of future tax subsidies decreases. The latter is the result of the limitation on the use of the Chèque Habitat in time.

Panel (b) in Figure A.14 plots the weighted average annual tax subsidy for Flanders (solid line) and Wallonia (dashed line) together with the difference (dotted line). The average subsidy in Flanders decreased by almost 35% from 1,388 euro in 2014 to 912 euro from 2015 onwards. In contrast, the average subsidy in Wallonia initially decreased slightly from 1,362 euro in 2014 to 1,220 euro in 2015, after which it increased by about 15% to 1,401 euro from 2016 onwards. The difference in the weighted average annual subsidies is about 30 euro over the pre-reform period. This is largely due to the one percentage point higher average marginal tax rate in Flanders. The difference decreased to nearly -500 euro after the regional reforms.



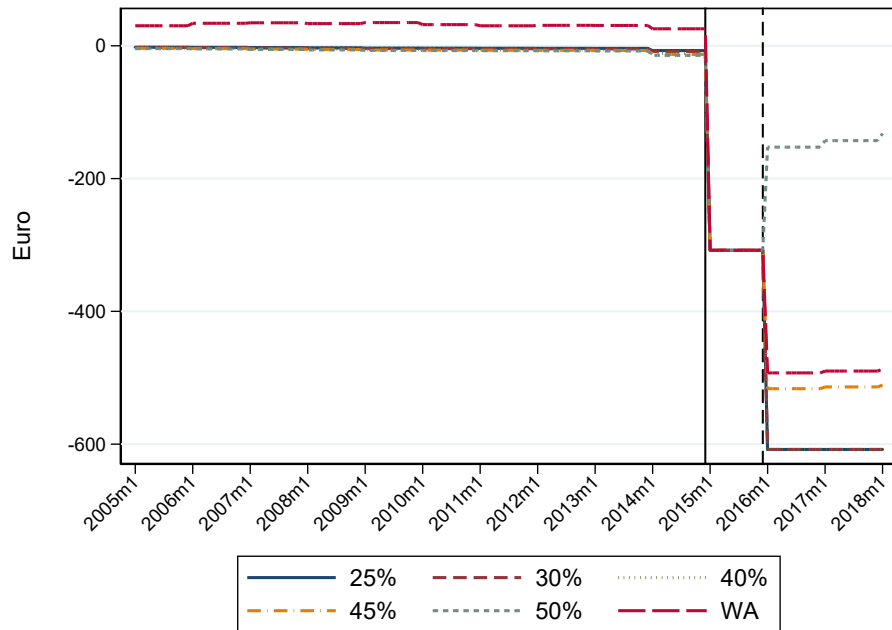
(a) Flanders



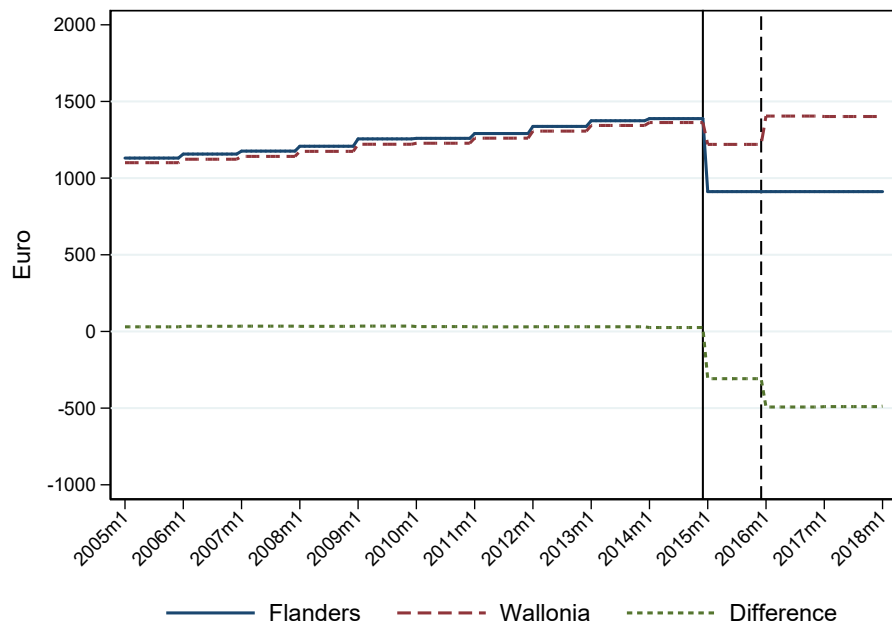
(b) Wallonia

Fig A.13. The annual tax subsidy for homeowners in Flanders and Wallonia together with the weighted average annual tax subsidy, 2005M1 to 2018M3

Notes: This Figure plots the size of the mortgage payment deduction and the Chèque Habitat in the first year of the mortgage per tax bracket for Flanders (a) and Wallonia (b). The solid and dashed vertical lines indicate December 2014 and December 2015 respectively. See Figure A.14 for the difference in the subsidy.



(a) The difference in the annual tax subsidy per tax bracket



(b) The weighted average annual tax subsidy in Flanders and Wallonia together with the difference

Fig A.14. The annual tax subsidy for homeowners

Notes: Panel (a) in this Figure plots the difference in the size of the mortgage payment deduction and the Chèque Habitat in the first year of the mortgage per tax bracket. Panel (b) plots the weighted average annual housing tax subsidy in the first year of the mortgage for Flanders (solid line) and Wallonia (dashed line) together with the difference (dotted line). The solid and dashed vertical lines indicate the last months before the regional reforms came into force, i.e. December 2014 and December 2015.

B Additional tables and figures for prices

Table B.1. Summary statistics full sample and repeat sample of sales

	Full sample		Repeat sales sample	
	Flanders (1)	Wallonia (2)	Flanders (3)	Wallonia (4)
A. single-family houses				
Price	235,706.27 (101,220.05)	169,191.29 (89,852.88)	218,093.13 (94,215.23)	167,859.20 (85,620.01)
Plot size (m ²)	555.62 (548.41)	629.57 (613.43)	412.99 (438.27)	554.86 (551.77)
Semi-detached	0.27 (0.45)	0.29 (0.45)	0.24 (0.43)	0.30 (0.46)
Detached	0.28 (0.45)	0.32 (0.47)	0.18 (0.39)	0.28 (0.45)
Square footage (m ²)	160.25 (53.61)	157.37 (53.25)	144.84 (49.52)	151.73 (52.22)
Habitable rooms (#)	5.62 (1.34)	5.60 (1.41)	5.42 (1.29)	5.52 (1.38)
Garages (#)	0.68 (0.58)	0.62 (0.60)	0.53 (0.57)	0.56 (0.59)
Age	64.85 (36.21)	86.53 (47.15)	73.97 (36.56)	91.68 (46.36)
Observations	355,798	192,256	71,165	27,904
B. Plots of undeveloped land				
Price	154,582.11 (79,452.25)	81,945.49 (60,138.12)		
Size (m ²)	671.16 (422.19)	1,076.30 (510.47)		
Price per m ²	306.17 (272.39)	96.95 (117.34)		
Observations	28,888	5,818		

Notes: This table reports the average sales price and the mean of the observed characteristics for single-family houses (Panel A) and plots of undeveloped land (Panel B) with the standard deviation in parentheses. The full samples contain all sales of respectively single-family houses and plots of undeveloped land in Flanders (1) and Wallonia (2), except for the outliers we removed as described in Section 4. The repeat sales sample contains all the single-family houses from the full sample for which we observe more than one sale.

Sources: sales data from GAPD, authors' calculations.

Table B.2. The difference in the average price and quality of single-family houses

	Full sample Fl. vs. Wa. (1)	Repeat sales Fl. vs. Wa. (2)	Flanders RS vs. Full (3)	Wallonia RS vs. Full (4)
Price (euro)	66,514.98 (0.00)	50,233.93 (0.00)	-17,613.14 (0.00)	-1,332.08 (0.00)
Lot size (in sq. m)	-73.95 (0.00)	-141.87 (0.00)	-142.63 (0.00)	-74.72 (0.00)
Semi-detached	-0.01 (0.00)	-0.06 (0.00)	-0.03 (0.00)	0.01 (0.00)
Detached	-0.04 (0.00)	-0.10 (0.00)	-0.10 (0.00)	-0.04 (0.00)
Square footage (in sq. m)	2.88 (0.00)	-6.89 (0.00)	-15.41 (0.00)	-5.64 (0.00)
Habitable rooms (#)	0.01 (0.00)	-0.10 (0.00)	-0.20 (0.00)	-0.09 (0.00)
Garages (#)	0.06 (0.00)	-0.03 (0.00)	-0.15 (0.00)	-0.06 (0.00)
Age	-21.68 (0.00)	-17.72 (0.00)	9.12 (0.00)	5.15 (0.00)

Notes: This table reports the difference in the average price and in the mean of the observed characteristics between the four different samples of sales of single-family houses. The p-value of a standard t-test is reported in parentheses. The average price and the mean of the observed characteristics are reported in Table B.1.

Sources: sales data from GAPD, authors' calculations.

Table B.3. The average quality of houses before and after the reforms

	Flanders				Wallonia			
	Pre (1)	Post (2)	Diff. (3)	Std. err. (4)	Pre (5)	Post (6)	Diff. (7)	Std. err. (8)
A. 2014Q4 vs. 2015Q1								
Lot size (m ²)	447.5	437.3	-10.2	15.5	573.7	560.8	-12.9	26.1
Semi-det.	0.25	0.26	0.01	0.01	0.31	0.30	-0.01	0.02
Detached	0.20	0.20	0.00	0.01	0.28	0.29	0.01	0.02
Sqr. ftg. (m ²)	147.12	148.05	0.93	1.65	152.66	153.95	1.30	2.43
Rooms (#)	5.44	5.44	-0.00	0.04	5.53	5.53	0.00	0.06
Garages (#)	0.50	0.58	0.08***	0.02	0.60	0.57	-0.03	0.03
Age	76.36	71.22	-5.14***	1.20	92.50	92.15	-0.35	2.17
B. Pre- vs. post-reform								
Plot size (m ²)	417.41	404.87	-12.54***	3.48	555.62	555.98	0.36	7.06
Semi-det.	0.24	0.24	0.01*	0.00	0.29	0.30	0.01	0.01
Detached	0.19	0.18	-0.01***	0.00	0.28	0.28	-0.01	0.01
Sqr. ftg. (m ²)	144.58	145.39	0.81**	0.39	151.59	152.10	0.51	0.67
Rooms (#)	5.42	5.41	-0.01	0.01	5.51	5.53	0.01	0.02
Garages (#)	0.54	0.53	-0.01***	0.00	0.56	0.57	0.01*	0.01
Age	72.31	77.41	5.10***	0.29	90.35	94.61	4.26***	0.59

Sources: sales data from GAPD, authors' calculations.

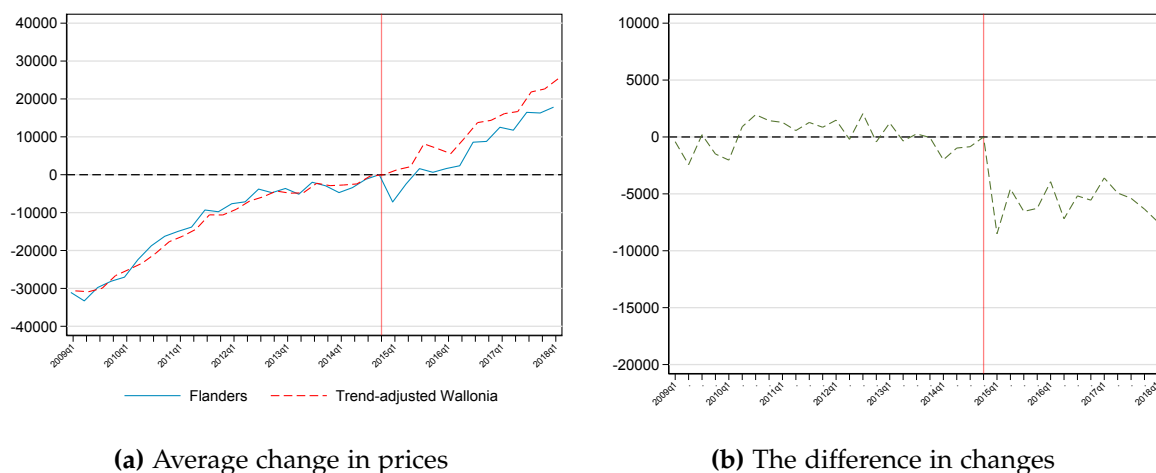
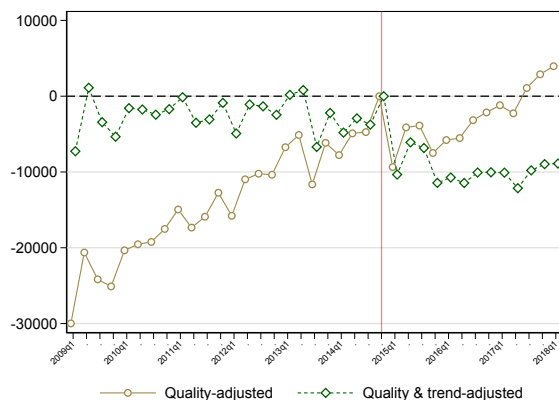


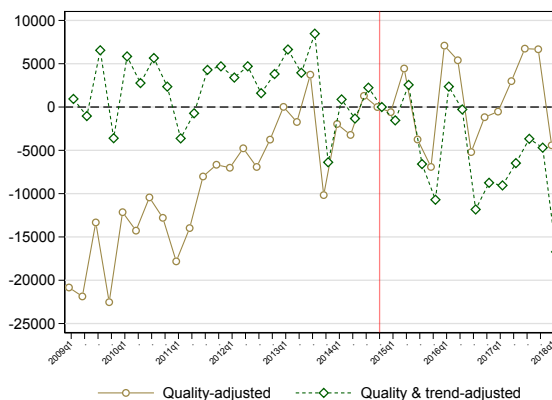
Fig B.1. The average change in quality-adjusted house prices and the difference in the average changes: the hedonic pricing method

Notes: Panel (a) plots the average change in quality-adjusted house prices for Flanders and the trend-adjusted average change in quality-adjusted house prices for Wallonia. The average change in quality-adjusted house prices is estimated using the hedonic pricing method in Equation 11. Panel (b) plots the estimates of β_t in Equation 13 using two series in panel (a) as dependent variable. The (red) vertical line indicates the reference period 2014Q4, the last quarter before the reforms.

Sources: sales data from GAPD, authors' calculations.



(a) Single-family houses



(b) Plots of undeveloped land

Fig B.2. The difference in the average change in quality-adjusted house and land prices with and without adjusting the changes for differential pre-trends

Notes: The figure plots the estimates of β_t in Equation 13 using the unadjusted and trend-adjusted average change in quality-adjusted prices of single-family houses (panel a) and plots of undeveloped land (panel b) as dependent variable. The (red) vertical line indicates 2014Q4, which is the baseline period.

Sources: sales data from GAPD, authors' calculations.

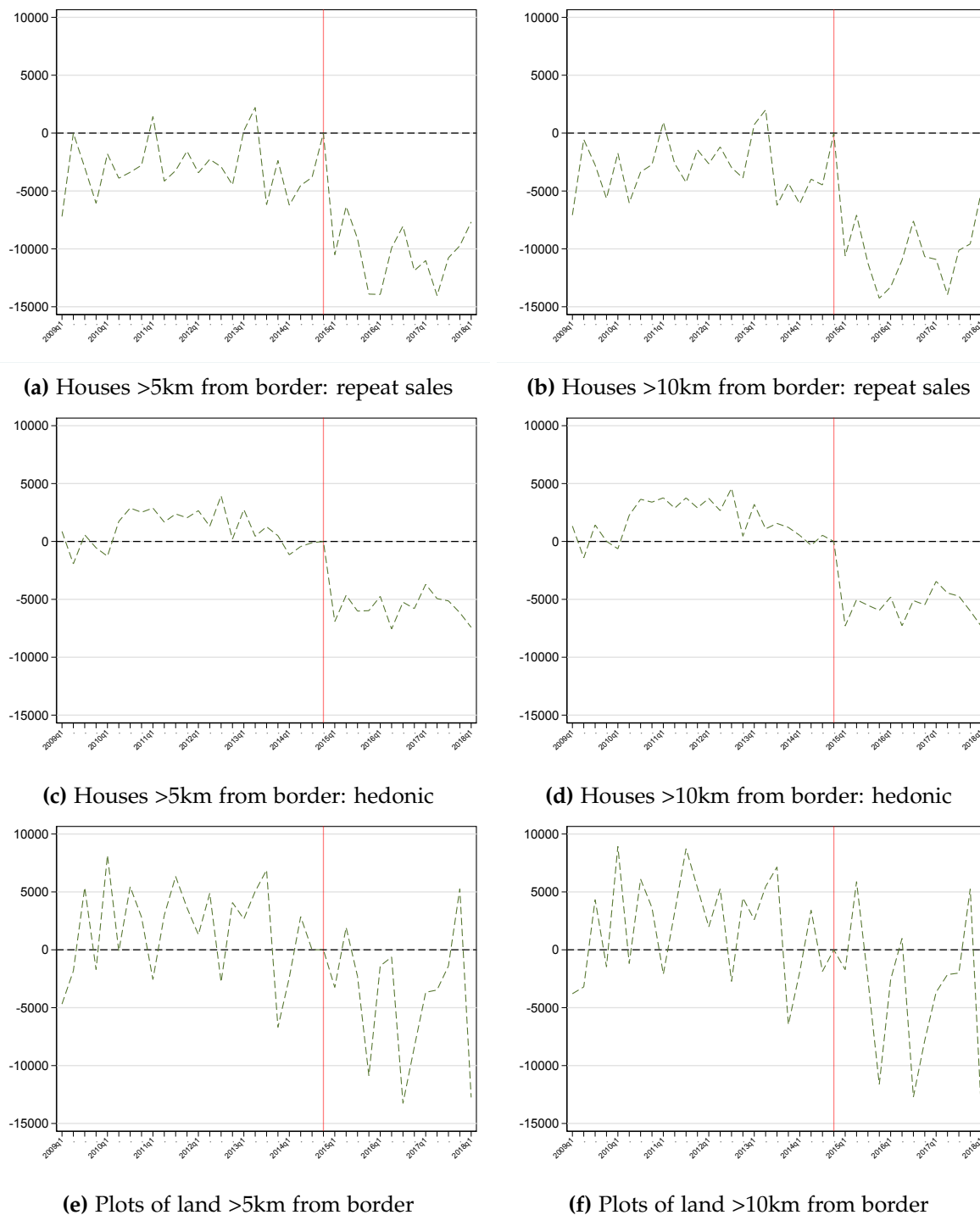


Fig B.3. The difference in the average changes in quality-adjusted prices: donuts

Notes: This figure plots the estimates of β_t in Equation 13 using the two ‘donut’ subsamples for single-family houses (panels a to d) and plots of undeveloped land (panels e and f). The (red) vertical line indicates the reference period 2014Q4, the last quarter before the reforms.

Sources: sales data from GAPD, authors’ calculations.

Table B.4. The average effect on house prices: the hedonic pricing method

	Repeat sales method			Hedonic pricing method		
	Full sample (1)	> 5km (2)	> 10km (3)	Full sample (4)	> 5km (5)	> 10km (6)
	A. Without observed characteristics			B. Full sample		
ATE(<i>d</i>)	-7,293*** (661.4)	-7,637*** (836.2)	-7,471*** (891.4)	-6,291*** (484.6)	-7,070*** (521.0)	-7,781*** (535.5)
ACR(<i>d</i>)	1.00*** (0.09)	1.05*** (0.12)	1.03*** (0.12)	0.86*** (0.07)	0.97*** (0.07)	1.07*** (0.07)
Obs.	74	74	74	74	74	74
Sales	99,069	90,087	81,751	99,069	90,087	81,751
Houses	47,671	43,399	39,388	47,671	43,399	39,388
	C. With observed characteristics			D. Repeat sales sample		
ATE(<i>d</i>)	-6,943*** (657.3)	-7,385*** (819.4)	-7,162*** (873.1)	-6,955*** (774.8)	-7,923*** (782.3)	-7,468*** (838.5)
ACR(<i>d</i>)	0.95*** (0.09)	1.01*** (0.11)	0.98*** (0.12)	0.96*** (0.11)	1.09*** (0.11)	1.03*** (0.12)
Obs.	74	74	74	74	74	74
Sales	548,054	495,977	451,070	99,069	90,087	81,751
Houses	496,656	449,289	408,707	47,671	43,399	39,388

Notes: The first row in this table presents estimates of β from the two-way fixed effects model in Equation 9 using the (trend-adjusted) average change in quality-adjusted house prices as the dependent variable ($\Delta \tilde{P}_{r,t}$). The average change in quality-adjusted house prices is estimated using four different methods: (A) the repeat sales method without observed characteristics, (B) the hedonic pricing method using the full sample, (C) the repeat sales method with observed characteristics, (D) the hedonic pricing method using the repeat sales sample. The average changes in quality-adjusted prices in Wallonia are adjusted for the difference in pre-reform trends by subtracting the estimate of region-specific linear trend. The vector $X_{i,t}$ in Equation 11 includes plot size, whether the house is detached, semi-detached or closed, square footage, the number of habitable rooms, garages and the age. The location fixed effects (α_l) capture cadastral sections. The third row reports the estimates of β divided by the net change in the difference in the average present value of the housing subsidies, which was 7,286.26 euro. The number of observations is 74. Robust standard errors in parentheses.

Sources: sales data from GAPD, authors' calculations.

Table B.5. Summary statistics for the different samples of sales of houses

	Repeat sales sample			Full sample		
	Full (1)	<5km (2)	<10km (3)	Full (4)	<5km (5)	<10km (6)
A. Flanders						
Price (euro)	218,069.09	219,684.01	221,340.89	235,706.27	236,678.46	237,486.70
Plot size (m ²)	413.23	408.18	410.48	555.62	549.81	538.23
Semi-detached	0.24	0.24	0.24	0.27	0.27	0.27
Detached	0.18	0.18	0.19	0.28	0.28	0.29
Square footage (m ²)	144.85	144.46	144.27	160.25	159.94	159.80
Habitable rooms (#)	5.41	5.41	5.42	5.62	5.61	5.61
Garages (#)	0.53	0.53	0.53	0.68	0.68	0.68
Age	74.01	73.36	72.77	64.85	64.26	63.73
B. Wallonia						
Price (euro)	167,847.35	159,039.05	154,627.68	169,191.29	159,818.75	155,268.96
Plot size (m ²)	555.73	549.41	538.23	629.57	623.05	614.22
Semi-detached	0.30	0.30	0.30	0.29	0.29	0.29
Detached	0.28	0.27	0.27	0.32	0.31	0.31
Square footage (m ²)	151.76	150.99	150.55	157.37	156.65	156.37
Habitable rooms (#)	5.52	5.51	5.51	5.60	5.61	5.61
Garages (#)	0.56	0.55	0.54	0.62	0.61	0.59
Age	91.74	93.26	94.29	86.53	88.31	89.61

Notes: This table reports the average price and characteristics of the different samples of sales of single-family houses.

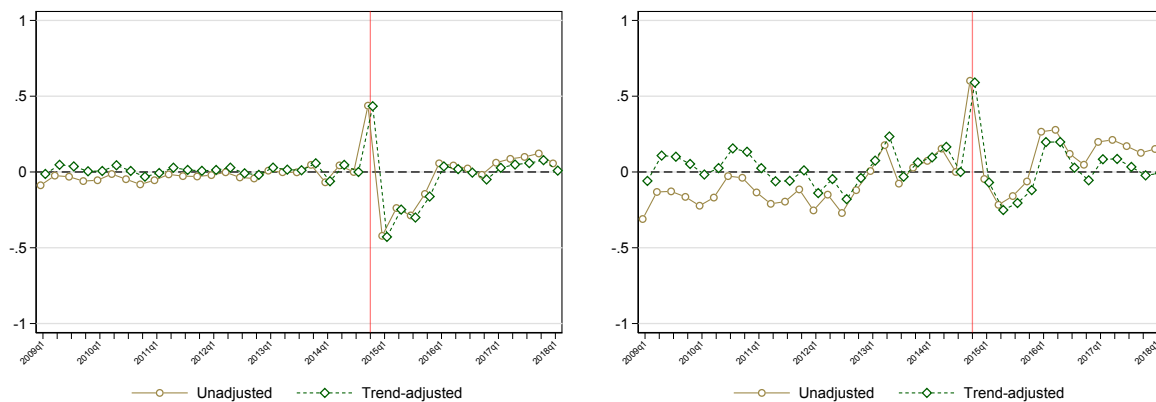
Sources: sales data from GAPD, authors' calculations.

C Additional tables and figures for sales and permits

Table C.1. Sales of undeveloped plots of land for residential development

	Deed (#)		Deed and land register (#)		Deed and land register (%)	
	F (1)	W (2)	F (3)	W (4)	F (5)	W (6)
2009	5708	1797	2614	657	0.46	0.37
2010	6517	1923	2898	675	0.44	0.35
2011	6244	1932	2681	692	0.43	0.36
2012	5042	1613	2374	570	0.47	0.35
2013	5059	1288	2350	474	0.46	0.37
2014	5515	1152	2578	471	0.47	0.41
2015	4495	1340	3364	721	0.75	0.54
2016	6010	1335	4751	817	0.79	0.61
2017	6492	1430	4947	905	0.76	0.63
2018Q1	1567	354	1192	206	0.76	0.58

Notes: The first two columns in this table presents the number of sales of undeveloped plots which the deed describes as land for residential development. Columns three and four report the number of sales of undeveloped plots which both the deed and the land register describe as land for residential development. Columns five and six presents the share of sales of undeveloped plots that is also described as land for residential development by the land register.



(a) single-family houses

(b) Plots of undeveloped land

Fig C.1. The difference in trends in sales: with and without adjustment for differential pre-trends

Notes: This figure plots the estimates of β_t in Equation 13 using the logarithm of sales of single-family houses (panel a) and plots of undeveloped land (panel b) as dependent variable with and without the adjustment for differential pre-trends. The baseline quarter is 2014Q3. The (red) vertical line indicates 2014Q4, the last quarter before the reforms.

Sources: sales data GAPD, authors' calculations.

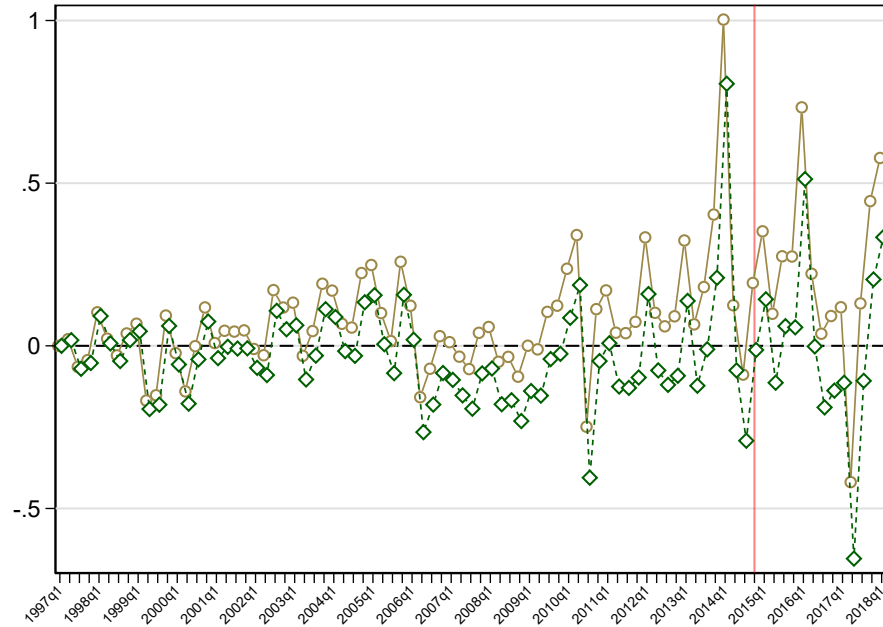


Fig C.2. The difference in trends in permits: with and without adjustment for differential pre-trends

Notes: This figure plots the estimates of β_t in Equation 13 using the logarithm of construction permits for single-family houses as dependent variable with and without the adjustment for differential pre-trends. The (red) vertical line indicates 2014Q4, which is the baseline quarter and the last quarter before the reforms of the housing subsidies.

Sources: permit data from Statbel, authors' calculations.